



Date: August,26, 2019

Exam Time: 2 hours

Final Exam

Summer semester 2018-2019

.Electrical circuits 1

Choose the correct answer.

* For the questions 1,2,3 consider the circuit shown in figure 1 :

1) The value of R_{th} with respect to terminals ab is:

.....
 2Ω

2) The value of V_{TH} from the terminals (a,b) is:

.....
 $6V$

3) The maximum power that can be delivered to the resistor R is:

..... $4.5 W$

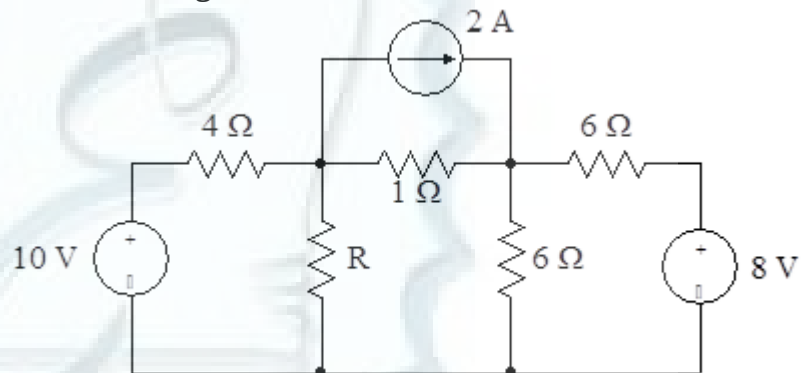


Fig.1

* For the questions 4,5,6,7 consider the circuit shown in figure 2, the switch S has been open for a long time. It is closed at $t=0$ s. and answer:

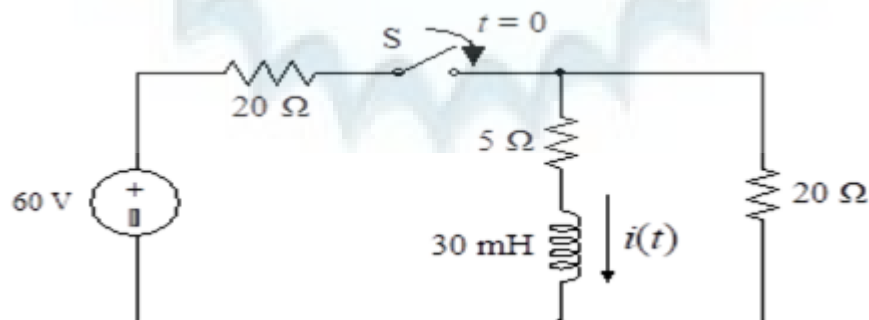


Fig.2

4) The value of $i(t = \infty)$ is:

.....2A.....

5) The current $i(t)$ for $t \geq 0$ is :

..... $i(t) = i(\infty) + [i(0^+) - i(\infty)] e^{-t/\tau} = 2 + (0 - 2)e^{-t/\tau} = 2(1 - e^{-t/\tau})$ A.....

6) The value of $V(t = 4\text{ms})$ across 5Ω resistor is:

$v(t) = 10 - 10e^{-(4/2)} = 8.65\text{V}$

7) The energy stored in the inductor at $(t = \infty)$ is:

.....60mJ.....

*If the voltage across a $5\text{-}\mu\text{F}$ capacitor is $v(t) = 10 \cos 314t$ V, answer 8,9:

8) The current through the capacitor is:

..... $-15.7 \sin 314t$ mA.....

9) The voltage at instant $t = 10\text{ ms}$ is:

.....-10V.....

* For the questions 10,11,12 consider the circuit shown in figure 3. Use super position theorem and answer:

10) The current I_x due to the 10V voltage source is:

a) **2 A** b) 1 A c) 500 mA d) none of all

11) The current I_x due to the 5A current source is:

a) 200 mA b) 2 A c) **-1 A** d) none of all

12) The total power delivered to 2Ω resistor is:

a) 1 W b) **2 W** c) 200 mW d) none of all

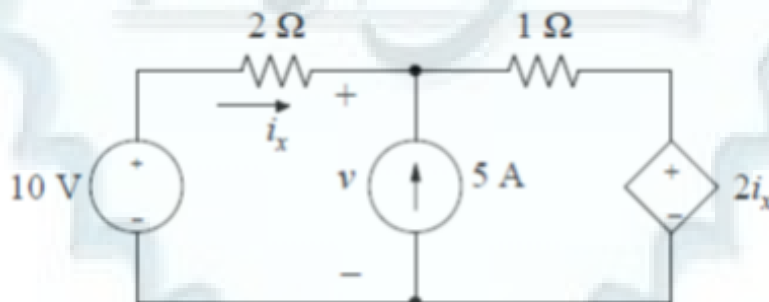
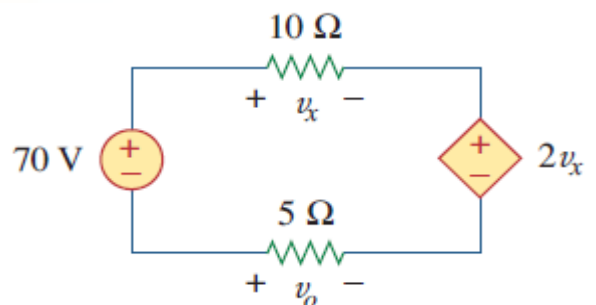


Fig.3

For the questions 13,14 consider the circuit shown in figure 4 :

13) The voltage v_o is:

a) 20 V b) **-10 V**
c) 10 V d) none of all



14) The current V_x is:

- a) **20 V** b) -10 V
c) 10 V d) none of all

Fig.4

15) The voltage across a 1.1 kW load produces a current of 10 A is :

- a) 11 kV b) **110 V** C) 1100kV d) none of all

16) The conductance of a light bulb with rated power of 100 W and rated voltage of 240 V is:

.....1.736 mS.....

17) The maximum current that a 2W, 80 k resistor can safely conduct is:

- a) 160 kA b) 40 kA c) **5 mA** d) none of all

* For the questions 18,19,20 ,21 (consider the circuit shown in figure 7.

18) The value of α (damping factor) is:

- a) **600 s⁻¹** b) 60 s⁻¹
c) 1000 s⁻¹ d) none of all

19) The value of ω_0 (resonant frequency) is:

- a) 100 s⁻¹ b) **1000 s⁻¹**
c) 600 s⁻¹ d) none of all

20) The general natural response $v(t)$ for $t \geq 0$ is:

- a) $i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$
c) **$i(t) = e^{-\alpha t} (B_1 \cos \omega_d t + B_2 \sin \omega_d t)$**

21) The energy dissipated in the resistor is:

- a) **5 J** b) 25 J
c) 500 mJ d) none of all

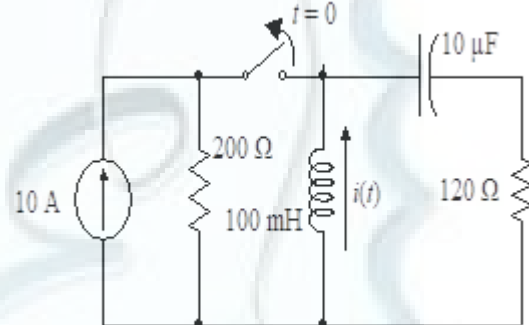


Fig.7

- b) $i(t) = e^{-\alpha t} (A_1 t + A_2)$
d) none of all

* For the questions 22,23,24,25 consider the circuit shown in figure 8 and answer:

22) The value of V_1 is:

.....20 V.....

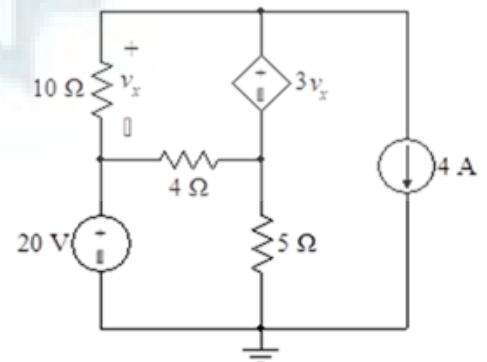
23) The value of the current (i_0) is:

.....-1A.....

24) The value of the voltage (V_2) is:

.....zero.....

25) The power absorbed by 4 ohm resistor is:



.....100W.....

Fig.8

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	25

Good Luck



CIRCUIT 1



فاينل

اسئلة سنوات



اتحاد هندسة الطاقة
والاتصالات

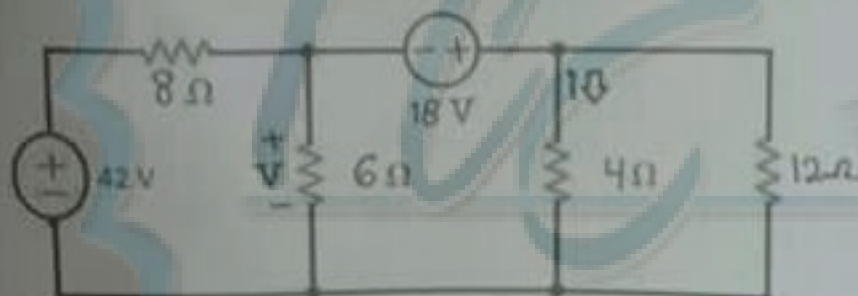
Electric Circuits – final Exam – May 20th, 2015

Q1: For the given circuit find:

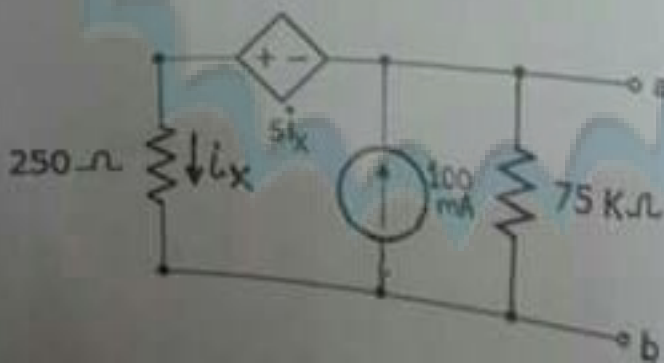
- 1) The equivalence resistance.
- 2) The power generated by the source.
- 3) The voltage across 5Ω resistor.



Q2: Using Nodal analysis, find V and i .

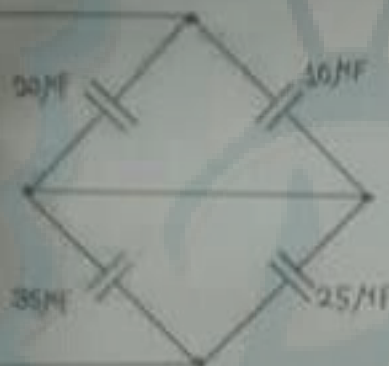


Q3: Answer the questions next page.



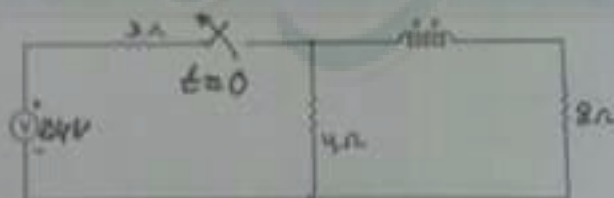
- 1) For the previous circuit, Determine the Thevenin and Norton equivalents for the given network.
- 2) Find the resistance that dissipate maximum power if it's connected between a and b terminals.

Q4: For the given circuits, calculate C_{eq} and L_{eq} .



Q5: For the given circuit below find the current in the inductor at:

- 1) $t < 0$ ($t < \text{zero}$).
- 2) $t = \infty$
- 3) $t = 2s$
- 4) The energy dissipated in 8Ω resistor.



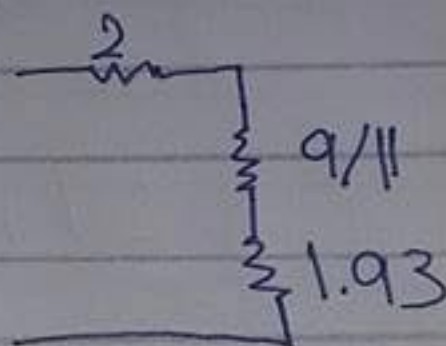
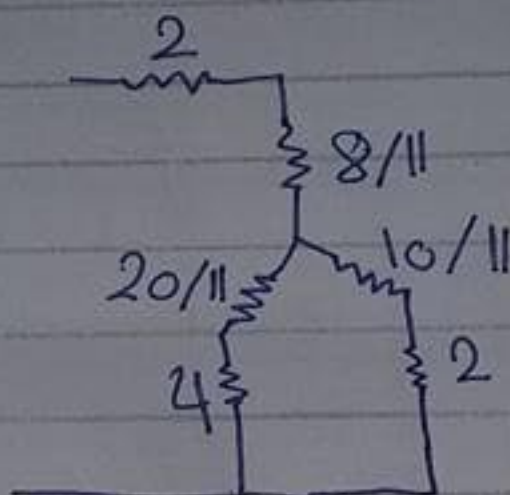
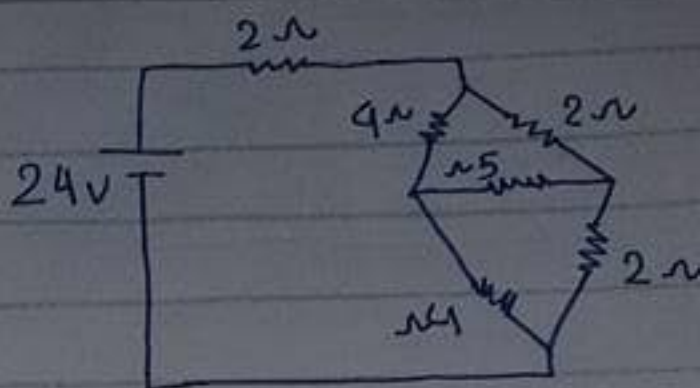
End of examinations papers.

Dr. Yousif Eltous --- Dr. Ibrahim Abu Harb

Final (1)

Question 1

(1)

Using $\Delta - Y$
Transformation

$$R_{eq} = 2 + \frac{8}{11} + 1.93$$

$$R_{eq} = 4.66 \Omega$$



L Final ①

$$\textcircled{2} \quad I_{\text{total}} = \frac{V}{R_{\text{eq}}} = \frac{24}{4.66}$$

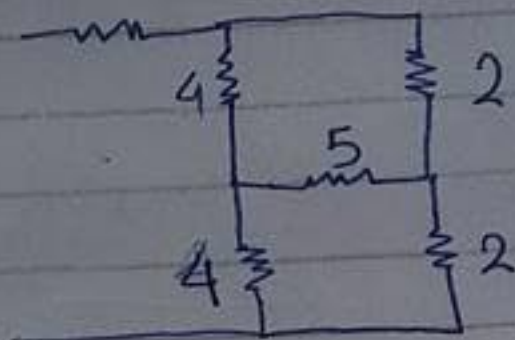
$$I_{\text{total}} = 5.15 \text{ A}$$

$$P = I \times V$$

$$P = 123.6 \text{ W}$$

③

* الجهد متوزع بالتساوي
على المقاومين 4Ω



* إذا مر تيار في تفرع التيار

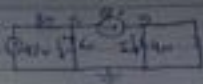
* و تفرع (التي) على المقاومين 4Ω

$$(V_{5\Omega} = \text{Zero})$$



Find ①

Current ②



• Supernode

$$\left[\frac{12}{2} V_1 - \frac{V_1}{6} + \frac{V_1}{4} + \frac{V_2}{3} \right] = 12$$

$$63 - 1.5V_1 = 2V_1 + 3V_2 + V_2$$

$$3.5V_1 + 4V_2 = 63 \quad \text{--- (1)}$$

$$\textcircled{2} V_2 - V_1 = 18 \quad \text{--- (2)}$$

$$\boxed{V_1 = -12 \text{ V}}$$

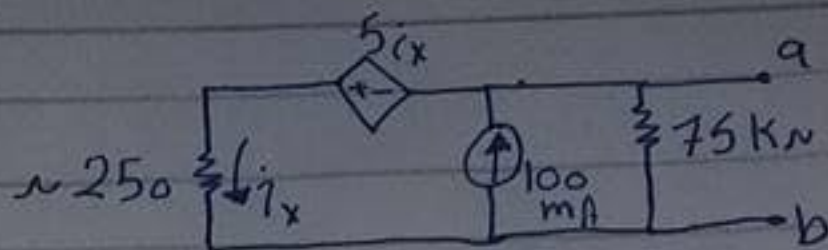
$$\boxed{V_2 = 16.8 \text{ V}}$$

$$I = \frac{V_2}{4\Omega}$$

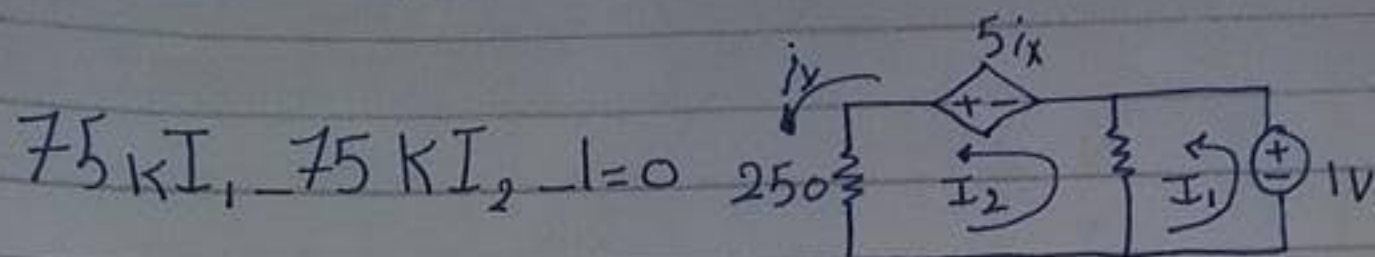
$$\boxed{I = 4.2 \text{ A}}$$

Final ①

Question ③ :-



$$\textcircled{1} R_{th} = R_N$$



$$75kI_1 - 75kI_2 - 1 = 0$$

$$75kI_1 - 75kI_2 = 1 \rightarrow \textcircled{1}$$

$$75 \cdot 250I_2 - 75kI_1 - 5i_x = 0$$

$$\boxed{i_v = I_2}$$

$$75245I_2 - 75kI_1 = 0 \rightarrow \textcircled{2}$$

$$I = 4.1 \text{ mA}$$

$$R_{th} = \frac{1V}{I_1}$$

$$\boxed{R_{th} = 244 \Omega}$$



Final ①

2 V_{th}

$$V_{th} = V(75 \text{ K}\Omega)$$

$$75 \text{ K}\Omega I_1 + 245 I_2 = 0 \rightarrow \textcircled{1}$$

$$\textcircled{*} 100 = I_2 - I_1 \rightarrow \textcircled{2}$$

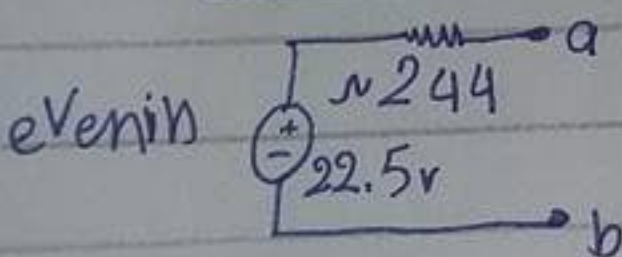
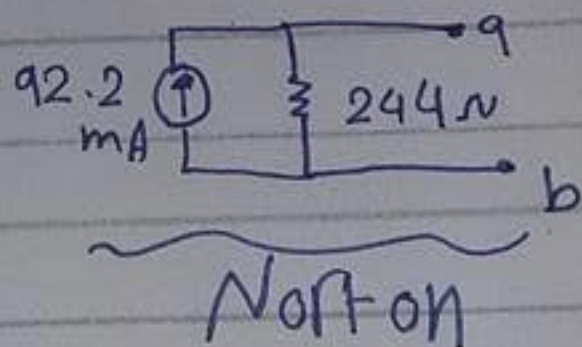
$$I_1 = -0.3 \text{ mA}$$

$$V_{th} = 0.3 \text{ m} \times 75 \text{ K}$$

$$V_{th} = 22.5 \text{ V}$$

$$\textcircled{3} I_N = V_{th} / R_{th}$$

$$I_N = 92.2 \text{ mA}$$

② At P_{max}

$$R_L = R_{th}$$

$$R_L = 244 \Omega$$

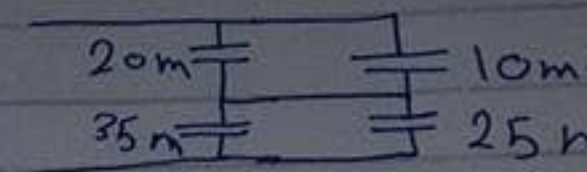
n't miss the change..
Be part of it !!

Final ①

Question (4):-

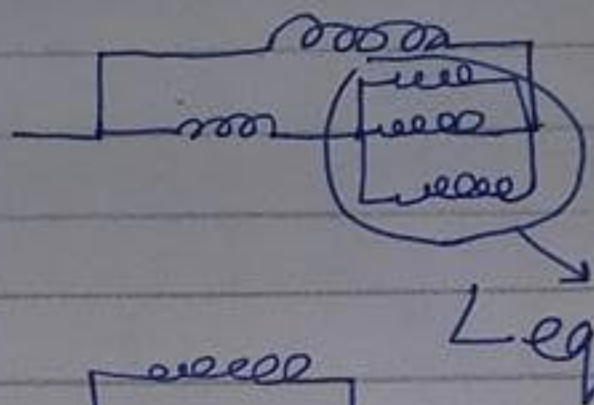
① $C_{eq} = \frac{30 \times 60}{30 + 60}$

$C_{eq} = 20 \mu F$



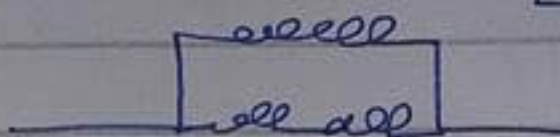
② $\frac{1}{L_{eq}} = \frac{1}{60m} + \frac{1}{30m} + \frac{1}{20m}$

$L_{eq1} = 10mH$



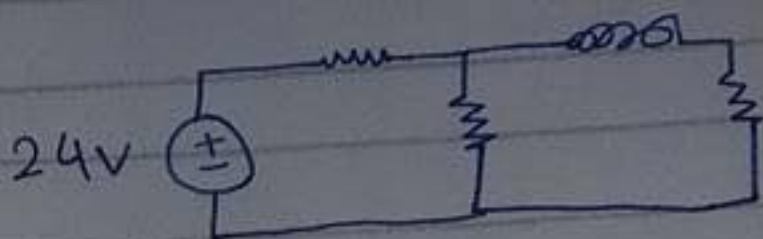
$L_{eq} = \frac{10m \times 35m}{10 + 35}$

$L_{eq} = 7.78mH$



Final (1)

Question (5)

① $t < 0$ $L \rightarrow S.C$ 

$$V. = \frac{(8/3)}{(8/3) + 3} * 24$$

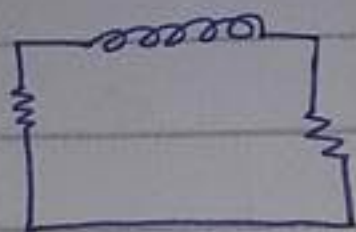
$$= 11.3 \text{ V}$$

$$I.(L) = \frac{V.}{8 \Omega}$$

$$\boxed{I. = 1.4 \text{ A}}$$

②

الحالة "تفرغ" *



$$\boxed{I_{\infty} = \text{Zero}}$$



Final (1)

$$\textcircled{3} \quad \tau = \frac{L}{R}$$

$$= \frac{1}{3} \text{ s}$$

$$i(t) = i \cdot e^{-t/\tau}$$

$$= 1.4 e^{-3t} \text{ A}$$

$$i(2t) = 3.47 \text{ mA}$$

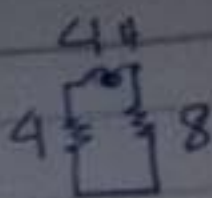
$$\textcircled{4} \quad \text{total energy} = \frac{1}{2} L i^2$$

$$= 3.92 \text{ J}$$

energy dissipated at 8 Ω

$$= \frac{8}{8+4} \times 3.92$$

$$= 2.62 \text{ J}$$





Q1: Find the voltage (v) for the network shown in Fig 1 using superposition theorem. (10marks)



Fig 1

Q2: For the circuit shown in Fig 2:

(10marks)

- Determine the value of R_1 to which a maximum power can be delivered.
- The maximum power delivered to R_1 .

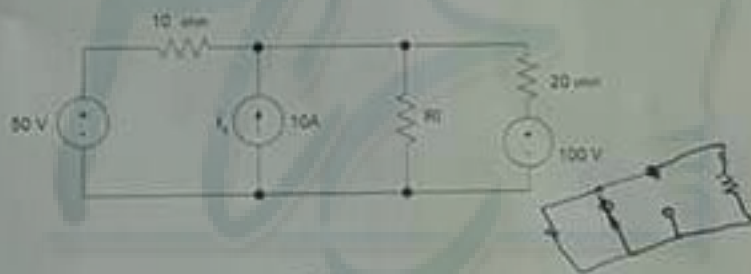


Fig 2

Q3: a) For the circuit shown in Fig 3 a find the equivalent capacitance (10 marks)

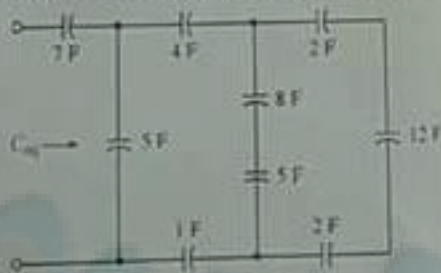


Fig 3a

- b) Calculate the power dissipated in the $40\ \Omega$ resistor and the voltage labeled V_C for the circuit given in fig 3.b

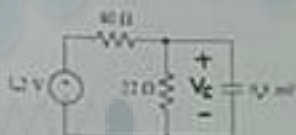


fig 3.b

- Q4: The voltage across a $33\ \text{mF}$ capacitor is shown graphically in Fig 4 (8 marks)

- a) Assuming the passive sign convention, sketch the resulting current waveform across the capacitor.

- b) Find the energy stored in the capacitor at $t=20\text{ms}$.

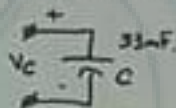


Fig 4

- Q5: For the circuit shown in Fig 5 the switch opens at $t=0$ after having been closed for a long time. Find:

- a) $i_1(t)$ for $t > 0^-$
 b) $i_2(t)$ for $t > 0^+$
 c) $V_2(t)$ for $t > 0^+$

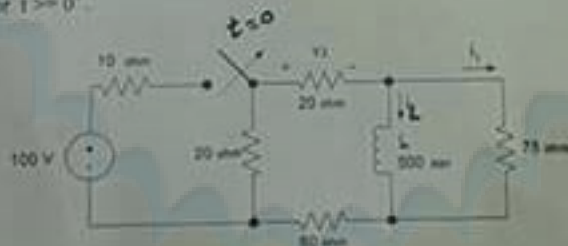


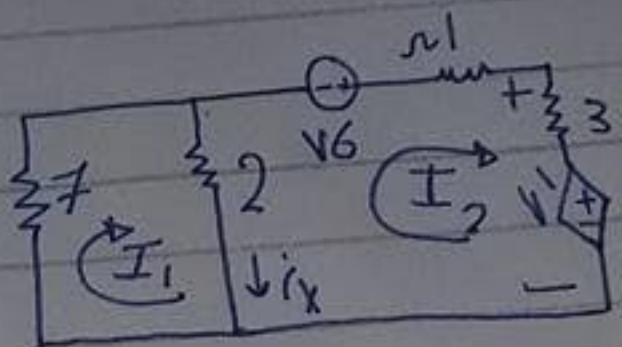
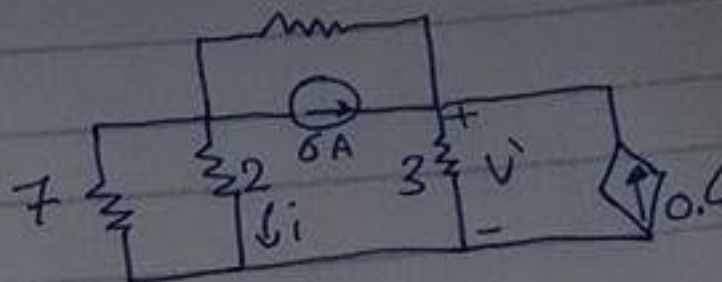
Fig 5

Good Luck

Final ②

Question ① :-

1) 6 A
4 V $\rightarrow$ S.C



Using mesh:

$$9I_1 - 2I_2 = 0 \rightarrow \text{①}$$

$$6I_2 - 2I_1 - 6 + 1.2(I_1 - I_2) = 0$$

$$-0.8I_1 + 4.8I_2 = 6 \rightarrow \text{②}$$

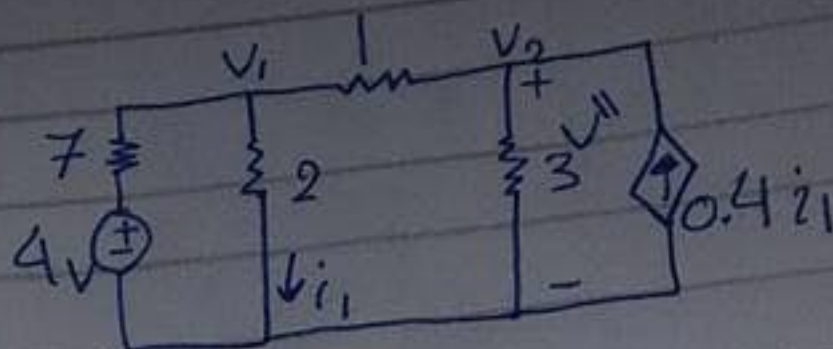
$$I_1 = 0.29 \text{ A}, I_2 = 1.3 \text{ A}$$

$$V = 3 \times 1.3 + 1.2(0.29 - 1.3)$$

$$\boxed{V = 2.688 \text{ V}}$$

Final(2)

[2] 4V



Using nodal :-

$$\left[\frac{4 - V_1}{7} = \frac{V_1}{2} + V_1 - V_2 \right] * 14$$

$$23V_1 - 14V_2 = 8 \rightarrow (1)$$

$$\left[V_1 - V_2 + \frac{0.4V_1}{2} = \frac{V_2}{3} \right] * 6$$

$$7 \cdot 2V_1 - 8V_2 = 0 \rightarrow (2)$$

$$V_2 = V'' = 0.69V$$

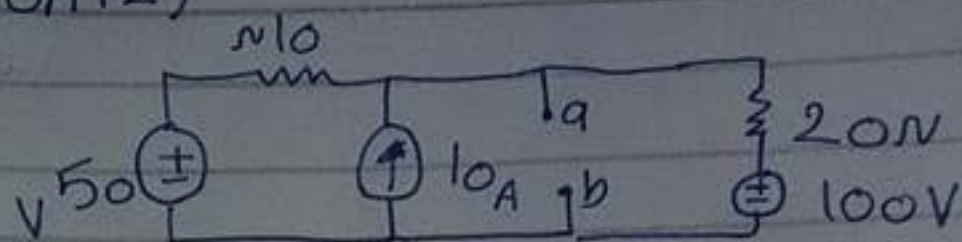
$$V = V' + V'' = 3.378V$$



Final (2)

Question (2)

(A)

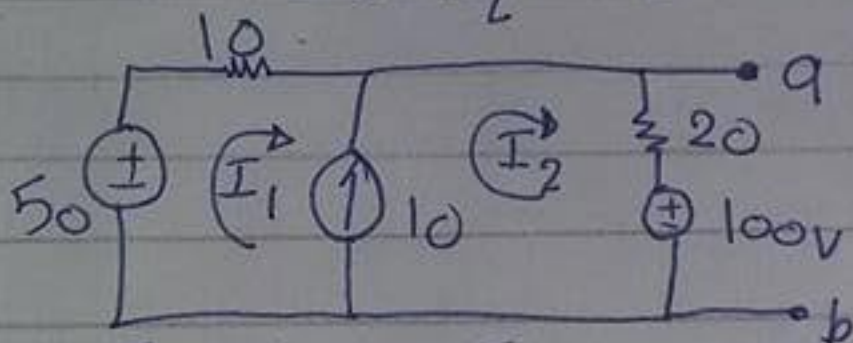
 $R_{th} :-$

$$R_{th} = 20 // 10$$

$$= 6.67 \Omega$$

$$R_L \Big|_{P_{max}} = R_{th}$$

$$\Rightarrow R_L = 6.67 \Omega$$

(B) $V_{th} :-$ 

$$10 I_1 + 20 I_2 + 100 - 50 = 0$$

$$I_1 + 2 I_2 = -5$$

$$I_2 - I_1 = 10$$

$$I_2 = 1.67 A$$

$$V_{th} = 100 + V_{(20\Omega)}$$

$$= 100 + 20 \times 1.67$$

$$= 133.4$$

$$* P_{max} = \frac{(V_{th})^2}{4 R_{th}}$$

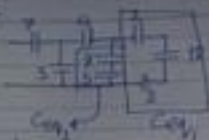
$$= \frac{(133.4)^2}{4 \times 6.67}$$

$$= 66 W$$

LEND

Question (3):

(A)



$$\frac{1}{C_{eq1}} = \frac{1}{3} + \frac{1}{5} + \frac{1}{12}$$

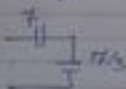
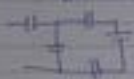
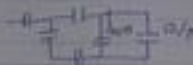
$$C_{eq1} = \frac{12}{13} \quad , \quad C_{eq2} = \frac{4}{13}$$

$$\frac{1}{C_{eq3}} = \frac{1}{4} + \frac{1}{4} + 1$$

$$= \frac{6}{4}$$

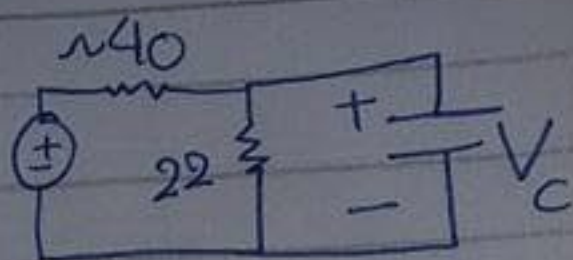
$$C_{eq3} = \frac{2}{3}$$

$$C_{eq3} = 3.13 F$$



Final ②

$$\textcircled{B} \quad I = \frac{1.2}{40 + 22} \\ = \frac{3}{155}$$



$$V(40\Omega) = \frac{40}{40 + 22} * 1.2 \\ = \frac{24}{31}$$

$$P(40\Omega) = I * V(40\Omega) \\ = 14.98 \text{ mW}$$

$$V_c = V(22\Omega) = \frac{22}{40 + 22} * 1.2 \\ = \boxed{0.42 \text{ V}}$$



Final(2)

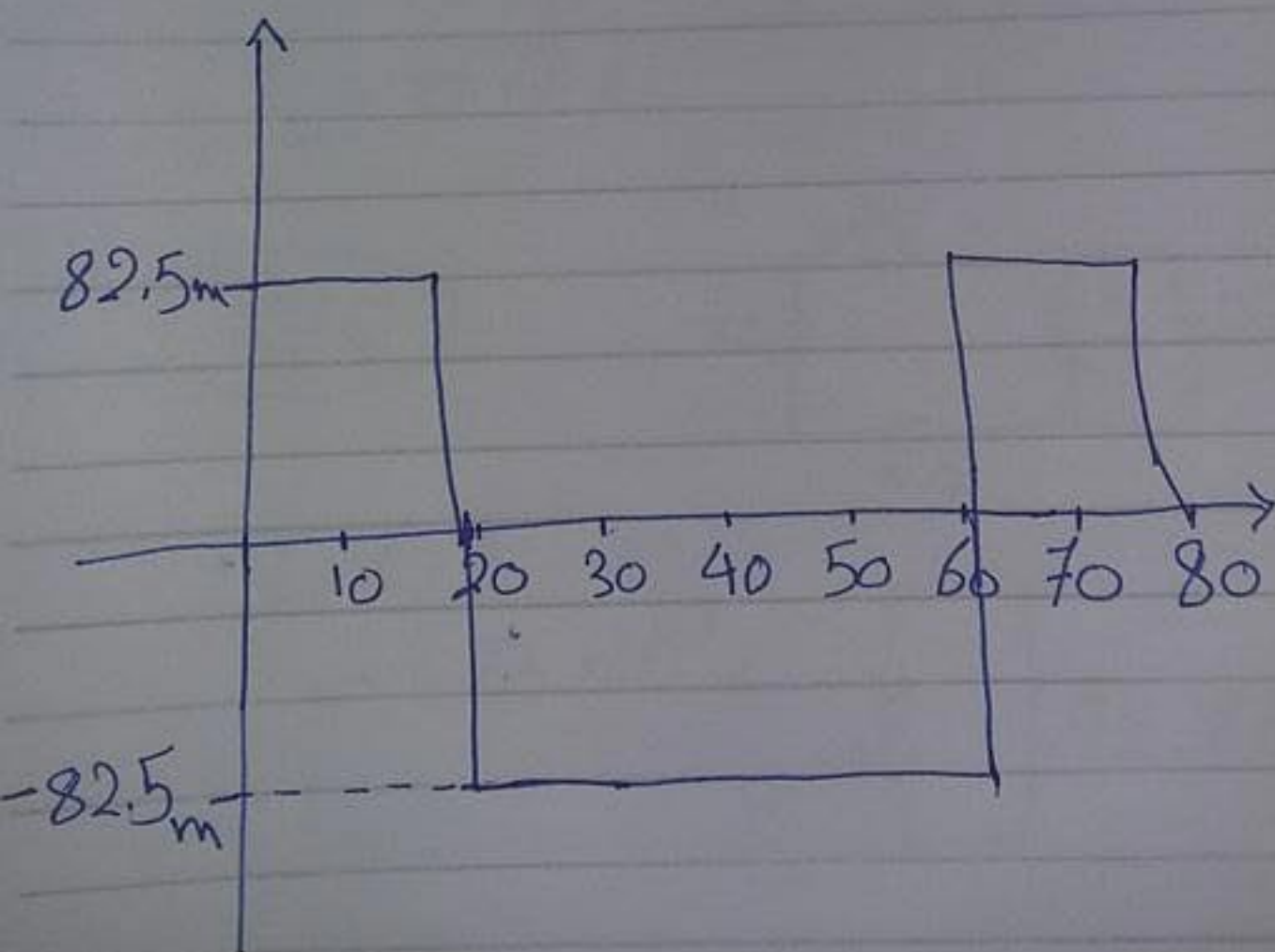
Question (4) :-

A

$$V(t) = \begin{cases} 2.5t & 0 < t < 20 \\ 100 - 2.5t & 20 < t < 60 \\ -200 + 2.5t & 60 < t < 80 \end{cases}$$

$$i(t) = C \frac{dV}{dt}$$

$$i(t) = \begin{cases} 82.5 \text{ mA} & 0 < t < 20 \\ -82.5 \text{ mA} & 20 < t < 60 \\ 82.5 \text{ mA} & 60 < t < 80 \end{cases}$$



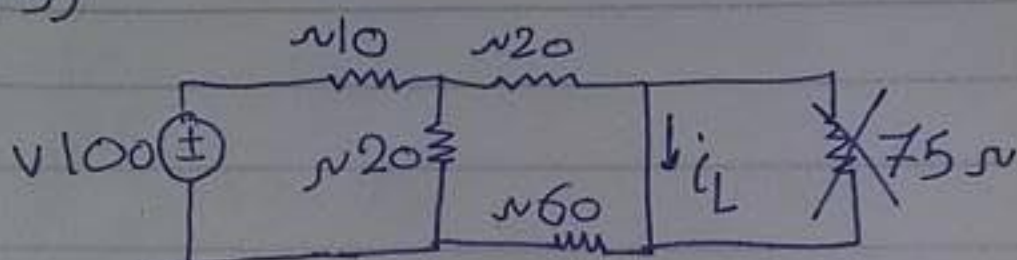
Final ②

B at 20 ms $\Rightarrow V = 50$ V

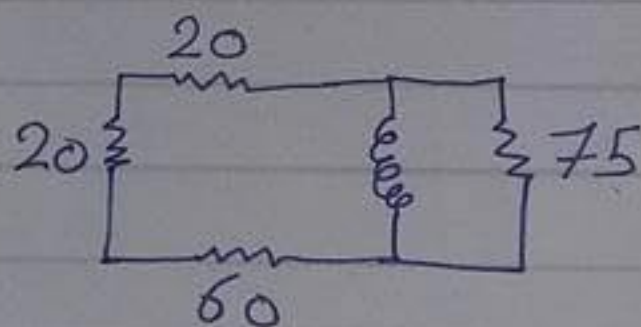
$$W = \frac{1}{2} C V^2$$

$$= 41.25 \text{ J}$$

Question (5)

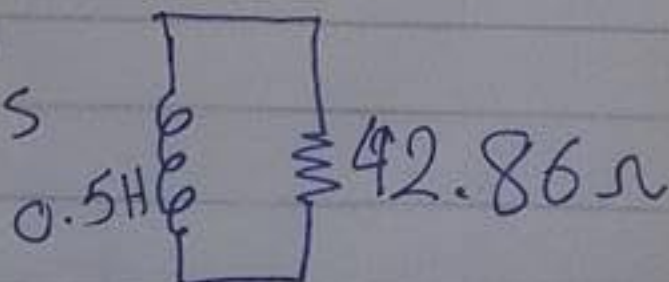
 $t < 0$ 

$$i_L = 0.77 \text{ A}$$

 $t > 0$ 

$$\tau = L/R$$

$$= 11.66 \times 10^{-3} \text{ s}$$





Electrical circuits 1.

Final Exam
First semester
2016-2017.

Date: Jan. 7, 2016

Exam Time: 2hours

Choose the correct answer.

* For the questions 1,2,3,4 consider the circuit shown in figure 1 and answer:

1) The value of V_0 is:

- a) 20 V b) 10 V
c) 40 V d) -40 V

2) The value of I is:

- a) 4A b) 1A
c) 2A d) -4A

3) The power absorbed by $5\ \Omega$ resistor is:

- a) 100W b) 40W c) 160W d) 80W

4) The power delivered by 40V voltage source is:

- a) 180W b) 40W c) 80W d) 100W

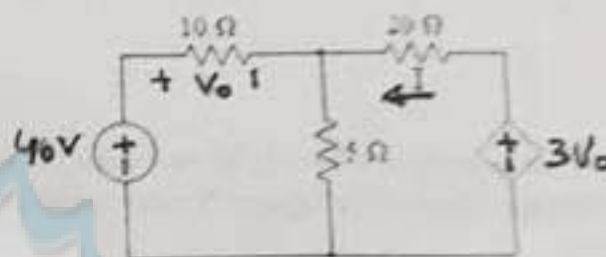


Fig.1

* For the questions 5,6,7,8 consider the circuit shown in figure 2 and answer:

5) The value of R_{TH} with respect to terminals (a,b) is:

- a) $6\ \Omega$ b) $3\ \Omega$
c) $14\ \Omega$ d) $16\ \Omega$

6) The value of V_{TH} from the terminals (a,b) is:

- a) 28V b) 12V
c) 32V d) 6V

7) The maximum power can be delivered to any load resistor is:

- a) 3W b) 9W c) 36W d) 18W

8) The current I is:

- a) 3A b) 1.5A c) 4.5A d) 9A

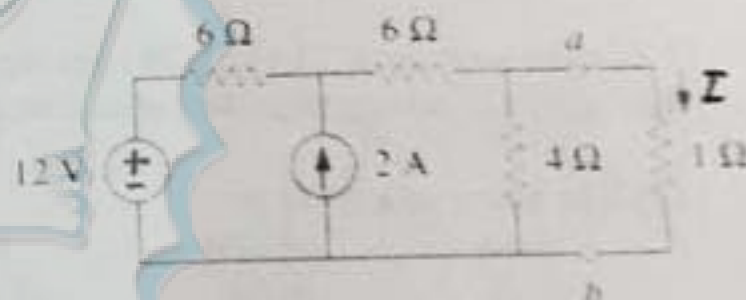


Fig.2

* For the questions 9,10,11,12 consider the circuit shown in figure 3 and answer:

9) The value of $v(0)$ is:

- a) 20V b) 24V
c) 8V d) 16V

10) the value of $v(t = \infty)$ is:

- a) 8V b) 0V
c) 24V d) 12V

11) The value of τ (time constant) is:

- a) 0.5s b) 0.5 ms
c) 2 ms d) 10 ms

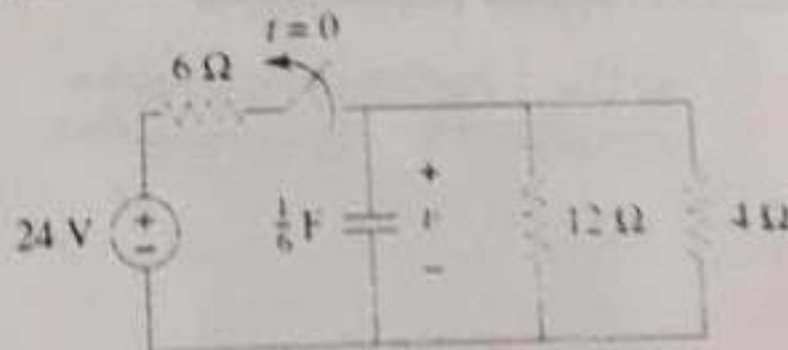


Fig.3

- 12) The value of v at $t = 750\text{ms}$ is: .
 a) 1.41V b) 0.8925 V c) 1.785V d) 3.57V

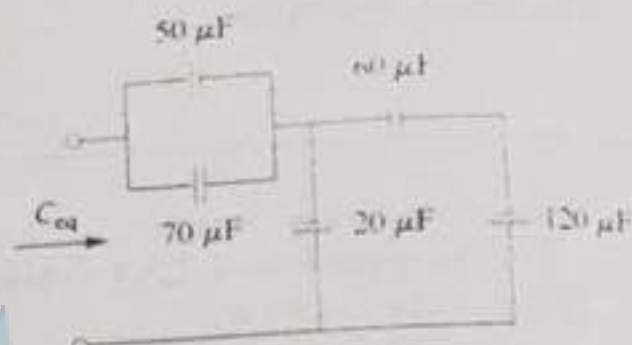


Fig 4

* For the questions 14,15,16 consider the circuit(the circuit under dc steady state conditions) shown in figure 5 and answer:

- 14) The inductor current i_L is:
 a) 2.5A b) 10A
 c) 5A d) 7.5A

- 15) The value of the capacitor voltage V_C is:
 a) 10V b) 20V
 c) 15V d) 5V

- 16) The energy stored in the capacitor is:
 a) 50 J b) 450 J c) 200 J d) 800 J

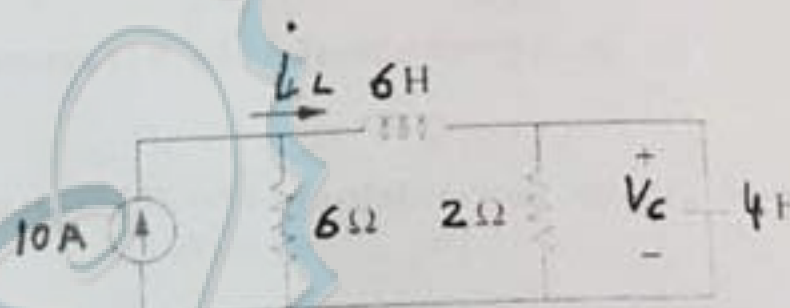


Fig.5

For the questions 17,18,19,20 consider the circuit shown in figure 6. The switch was open for a long time, but closed at $t=0$, If $i(0)=10\text{A}$. Answer:

- 17) The value of $i(t=\infty)$ is:
 a) 6A b) 12A
 c) 1A d) 5A

- 18) The time constant is:
 a) 500ms b) 400ms
 c) 200ms d) 20 ms

- 19) The value of $i(t=100\text{ms})$ is:
 a) 3A b) 6A
 c) 4.2A d) 8.4A

- 20) The complete response $i(t)$ is:

- a) $i(t)=12+6e^{-50t}\text{ A}$ b) $i(t)=6+4e^{-5t}\text{ A}$
 c) $i(t)=6+e^{-5t}\text{ A}$ d) $i(t)=6+4e^{-0.5t}\text{ A}$

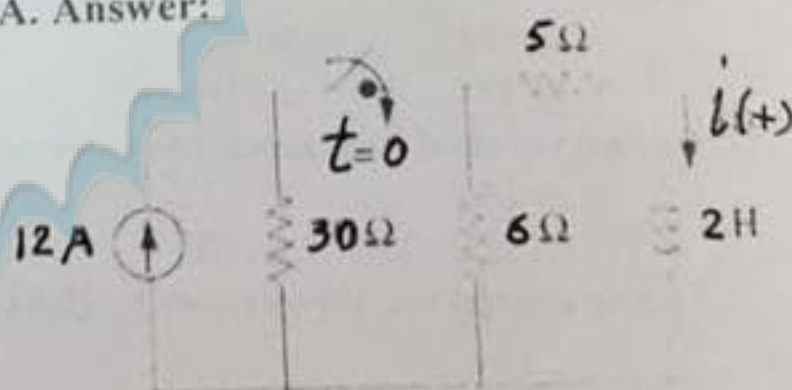


Fig.6

* For the questions 21,22,23,24,25 consider the circuit shown in figure 7. Assuming $v(0)=5V, i(0)=0, L=1H, C=10mF, R=1.923 \Omega$. Answer:

21) The value of α (damping factor) is:

- a) 52 b) 6.5
c) 13 d) 26

22) The value of ω_o (resonant frequency) is:

- a) 40 b) 10
c) 20 d) 5

23) The value of $S1$ is:

- a) -6 b) -4 c) -2 d) -25

24) The value of $S2$ is:

- a) -100 b) -25 c) -12.5 d) -50

25) The natural response $v(t)$ for $t > 0$ is:

- a) $v(t) = -0.2083e^{-2t} + 5.208e^{-50t}$ b) $v(t) = -0.2083e^{-4t} + 5.208e^{-25t}$
c) $v(t) = 0.2083e^{-2t} - 5.208e^{-50t}$ d) $v(t) = 0.2083e^{-4t} - 5.208e^{-25t}$

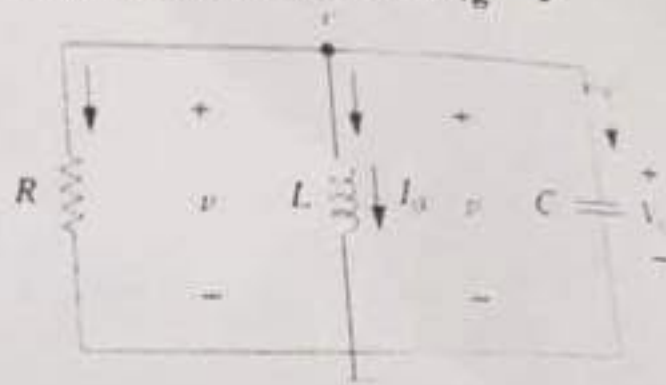


Fig.7

1	2	3	4	5	6	7	8	9	10	11	12	13
14	15	16	17	18	19	20	21	22	23	24	25	

Good Luck

a) -6

b) -4

c) -2

d) -25

24) The value of S_2 is:

a) -100

b) -25

c) -12.5

d) -50

25) The natural response $v(t)$ for $t > 0$ is:

a) $v(t) = -0.2083e^{-2t} + 5.208e^{-50t}$

b) $v(t) = -0.2083e^{-4t} + 5.208e^{-25t}$

c) $v(t) = 0.2083e^{-2t} - 5.208e^{-50t}$

d) $v(t) = 0.2083e^{-4t} - 5.208e^{-25t}$

Fig.7

1	2	3	4	5	6	7	8	9	10	11	12	13
a	c	b	c	b	d	a	b	c	b	a	c	b
14	15	16	17	18	19	20	21	22	23	24	25	
d	c	b	a	c	d	b	d	d	c	d	c	

b

Good Luck

CASIO

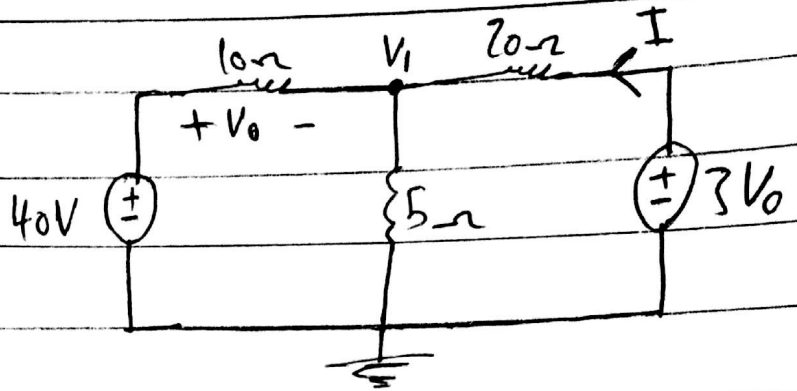
fx-991EX

CLASSWIZ

f(x)

①

Q1 → find V_o , I , power for 5Ω and power for $40V$ voltage source →



Soln

Using nodal method →

$$\frac{V_1 - 40}{10} + \frac{V_1}{5} + \frac{V_1 - 3V_o}{20} = 0 \rightarrow * V_o = 40 - V_1$$

$$\therefore \frac{V_1 - 40}{10} + \frac{V_1}{5} + \frac{4V_1 - 120}{20} \times 20$$

$$2V_1 - 80 + 4V_1 + 4V_1 - 120 = 0 \rightarrow \therefore \boxed{V_1 = 20V}$$

$$V_o = 40 - V_1 = \boxed{20V}$$

$$I = \frac{3V_o - V_1}{20} = \frac{3 \times 20 - 20}{20} = \boxed{2A}$$

$$P_5 = I_5 V_5 \quad I_5 = \frac{V_1}{5} = 4A \quad V_5 = V_1 = 20V$$

$$\therefore P_5 = 4 \times 20 = \boxed{80W}$$

$$P_{40V} = I_{10\Omega} \times 40 \quad I_{10\Omega} = \frac{40 - V_1}{10} = 2A$$

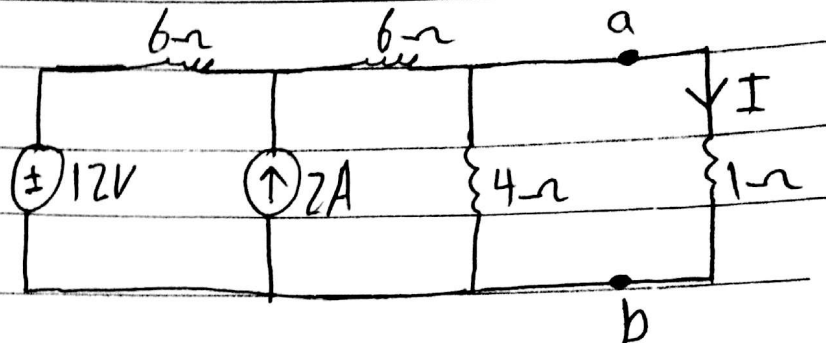
$$\therefore P_{40V} = 2 \times 40 = \boxed{80W}$$

Don't miss the change..
Be part of it !!



②

Q2 → Find R_{th} , V_{th} ,
I and maximum power
can be delivered to
any load resistor →

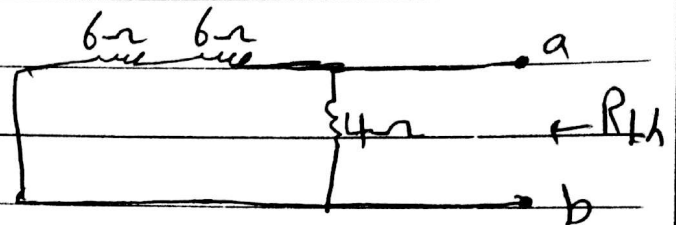


Soln

* to find R_{th} →

6 series with 6 = 12Ω

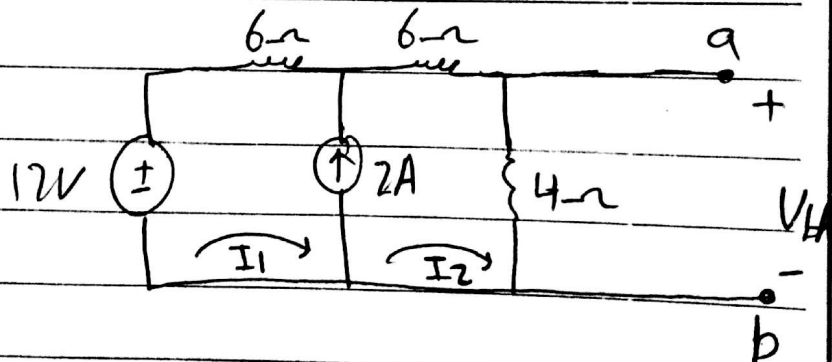
$$12 // 4 = 3\Omega$$



* to find V_{th} →

using mesh method →

$$I_2 - I_1 = 2A$$



$$-12 + 6I_1 + 10I_2 = 0$$

$$6I_1 + 10I_2 = 12$$

$$\therefore I_1 = -0.5A$$

$$I_2 = 1.5A$$

$$V_{th} = 4I_2 = 4 \times 1.5 = 6V$$

2.5



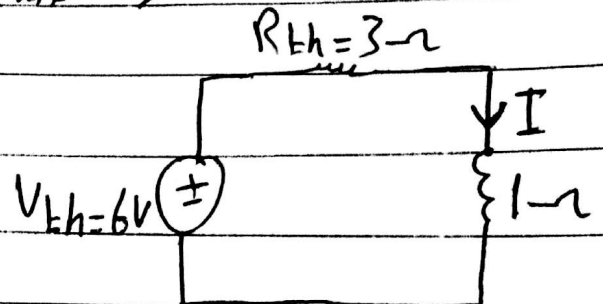
(3)

شكلا السؤال الثاني

* The equivalent thevenin circuit →

$$-6 + 3I + 1I = 0$$

$$\therefore I = \frac{6}{4} = 1.5A$$

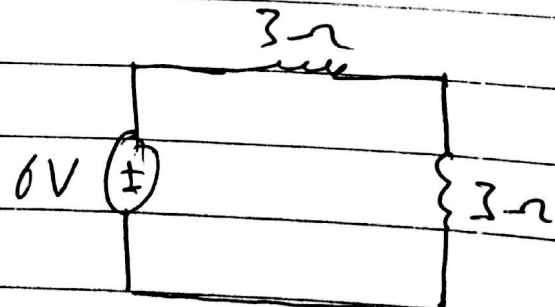


* ~~power~~ maximum power to any load resistor
(تعني لأي د ولأي د 1Ω)

$$R_L = R_{th} = 3\Omega$$

* إذا كانا متساوي

$$\therefore P = \frac{V_{th}^2}{4R_{th}} = \frac{6^2}{4 \times 3} = 3W$$

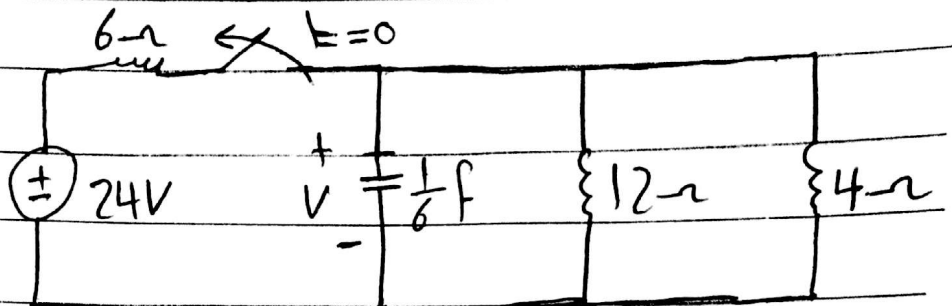


④

Q3 → find $V(t)$

V at $t = \infty$, T

and V at $t = 750\text{ms}$



Sol:

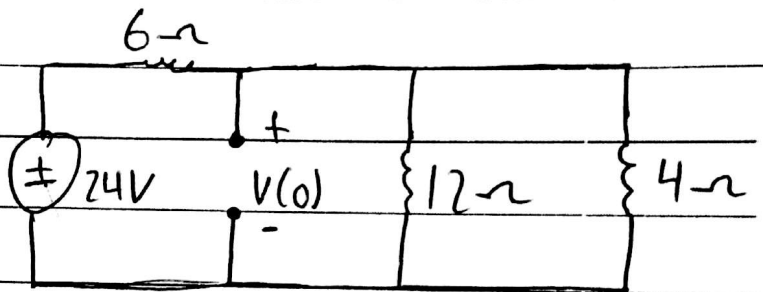
$V(0) \rightarrow$ at $t < 0$

$$12//4 = 3\Omega$$

Voltage divider →

$$V(0) = V_{3\Omega}$$

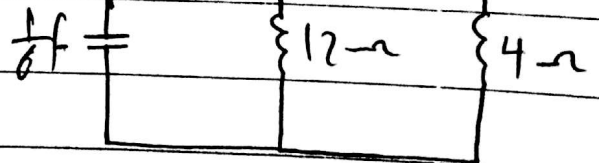
$$= \frac{3}{3+6} \times 24 = \boxed{8V}$$



at $t > 0 \rightarrow$

$$R_{eq} = R_{th} = 12//4 = 3\Omega$$

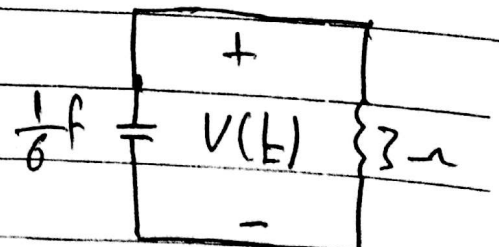
$$\tau = R_{eq} \times C = 3 \times \frac{1}{6} = \boxed{0.5s}$$



$$V(t) = V_0 e^{-\frac{t}{\tau}} = 8e^{-2t}$$

at $t = \infty \rightarrow$

$$V = 8e^{-2 \times \infty} = 8 \times \frac{1}{e^{\infty}} = 8 \times 0 = \boxed{0}$$



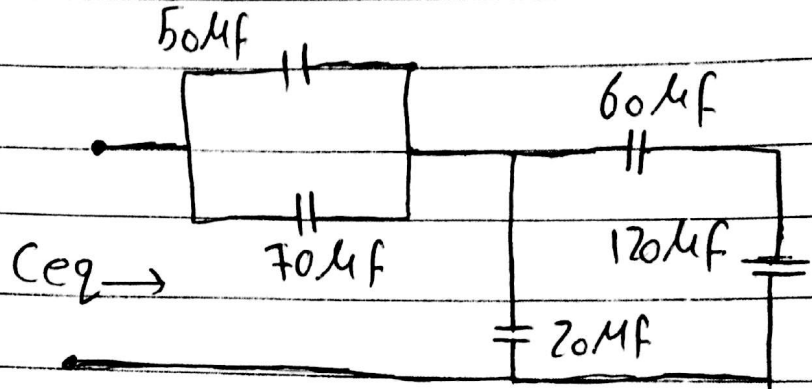
at $t = 750\text{ms} \rightarrow$

$$V = 8e^{-2 \times 0.75} = \boxed{1.785V}$$

Don't miss the change..
Be part of it !!



⑤

Q4 → find C_{eq} →

soln

$$50\text{M} \parallel 70\text{M} = 120\text{Mf}$$

$$60\text{M} \text{ series } 120\text{M} = 40\text{Mf}$$

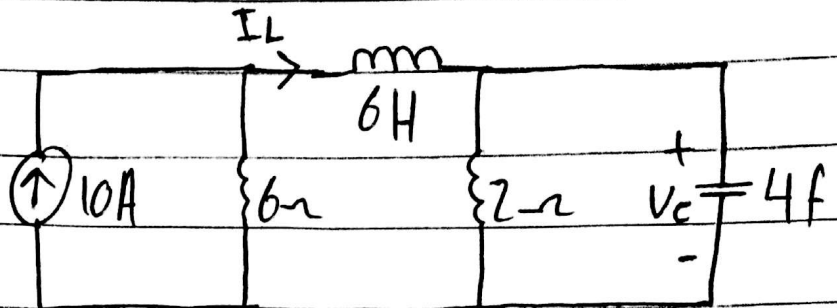
$$20\text{M} \parallel 40\text{M} = 60\text{Mf}$$

$$120\text{M} \text{ series with } 60\text{M} = 40\text{M} = C_{eq}$$



⑥

Q5 → In circuit
under dc steady
state conditions
find I_L , V_C and
energy stored in
the capacitor →

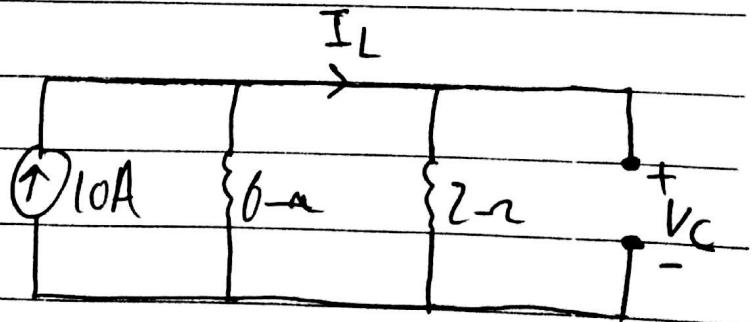


soln

under dc steady state conditions,

using current divider →

$$I_L = \frac{6}{2+6} \times 10 = \boxed{7.5A}$$



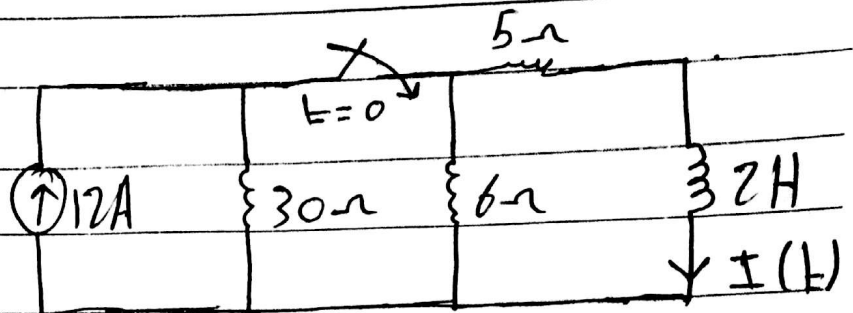
$$V_C = V_{2\Omega} = 2I_L = \boxed{15V}$$

$$W_C = \frac{1}{2} C V^2 = \frac{1}{2} \times 4 \times 15^2 = \boxed{450J}$$



7

Q6 → the switch
was open for along
time but closed at
 $t=0$, if $i(0) = 10A$
find $i(t=\infty)$, τ ,
 $i(t=100ms)$ and $i(t)$



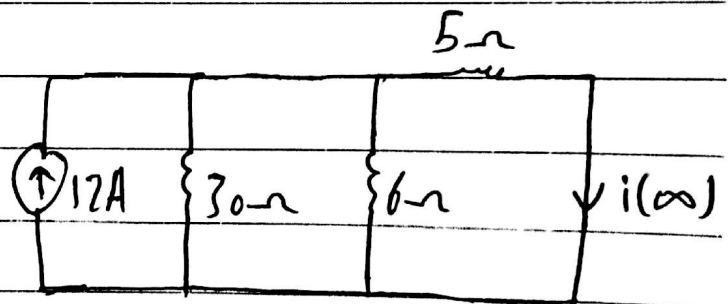
soln (step response)

$i(t=\infty) \rightarrow$

$$30 // 6 = 5 \Omega$$

using current divider $\rightarrow$

$$i(\infty) = i_{5\Omega} = \frac{5}{5+5} \times 12 = 6A$$

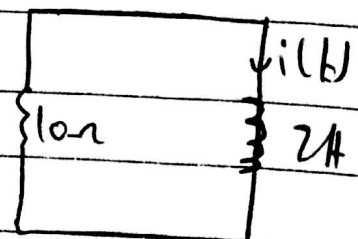


$$R_{eq} = 30 // 6 + 5 = 10 \Omega$$

$$\tau = \frac{L}{R_{eq}} = \frac{2}{10} = 200ms$$

$$i(t) = i(\infty) + (i(0) - i(\infty))e^{-\frac{t}{\tau}}$$

$$\therefore i(t) = 6 + 4e^{-5t} A$$



$$i(t=100ms) = 6 + 4e^{-5 \times 0.1} = 8.43 A = 8.4 A$$

Don't miss the change..
Be part of it !!

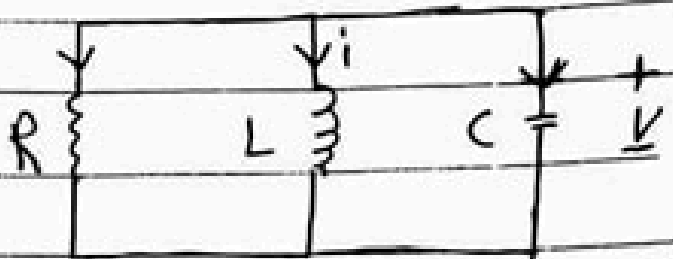


⑧

Q7) $V(0) = 5V$, $i(0) = 0$ $L = 1H$ $C = 10mF$ $R = 1.923\Omega$

Soln

(source free parallel RLC)



$$\alpha = \frac{1}{2RC} = 26$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = 10$$

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} = -2$$

$$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2} = -50$$

$\alpha > \omega_0$ (over damped)

$$\therefore V(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

$$V(0) = A_1 e^{0 \times t} + A_2 e^{0 \times t} = 5 \rightarrow \therefore \boxed{A_1 + A_2 = 5} \quad (1)$$

$$i_R + i_L + i_C = 0 \rightarrow \frac{V(t)}{R} + i(t) + C \frac{\partial V(t)}{\partial t} = 0$$

$$\therefore \frac{V(0)}{R} + i(0) + C \frac{\partial V(0)}{\partial t} = 0 \rightarrow \frac{5}{1.923} + 0 + 10m \frac{\partial V(0)}{\partial t} = 0$$

$$\therefore \frac{\partial V(0)}{\partial t} = -260$$



⑨

* الآلة تشتق صدارة الجهد الأولية ونفوض فيها $t=0$
ونسا فيها ب -260

$$V(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

$$\frac{\partial V(t)}{\partial t} = A_1 s_1 e^{s_1 t} + A_2 s_2 e^{s_2 t}$$

$$\frac{\partial V(0)}{\partial t} = A_1 s_1 e^{s_1 \times 0} + A_2 s_2 e^{s_2 \times 0} = -260$$

$$\therefore A_1 \times -2 + A_2 \times -50 = -260 \quad / -2$$

$$\therefore \boxed{A_1 + 25A_2 = 130} \quad (2)$$

$$\therefore A_1 = -0.2083 \quad A_2 = 5.208$$

$$\therefore V(t) = -0.2083 e^{-2t} + 5.208 e^{-50t}$$





Q1. Use superposition theorem to find i_o and the power absorbed by the 6Ω resistor in the circuit shown in figure 1.

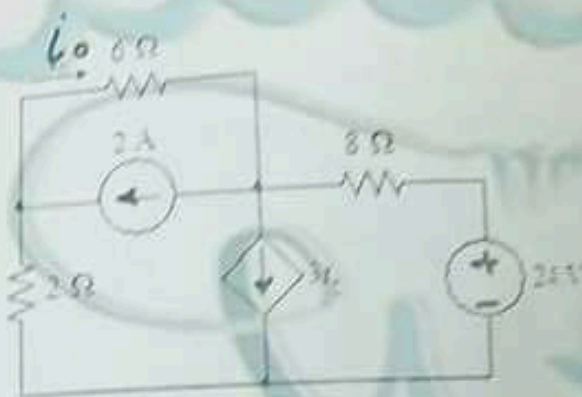


Fig.1

Q2. In the circuit of Figure 2,

- Find R so that maximum power is transferred to the resistance R .
- Find this maximum power.

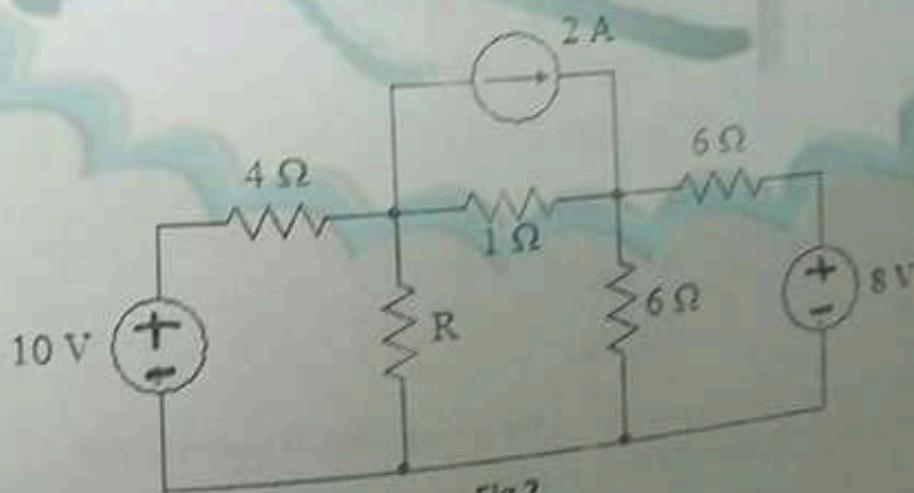


Fig 2

* For the questions 1,2,3 consider the circuit shown in figure 3 under dc steady state conditions and answer:

- 1) The capacitor voltage is:
a. 10V b. 5V c. 20V d. 60V
- 2) The energy stored in the capacitor is:
a. 10J b. 0.5J c. 90J d. 40J
- 3) The current i_x is:
a. 12A b. 4A c. 8A d. 3A

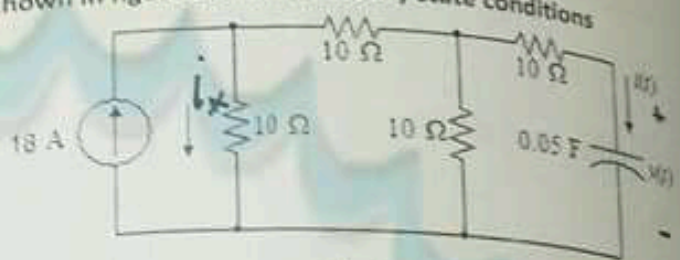


Fig 3

* For the questions 4,5,6 consider the circuit shown in figure 4. Use Norton's theorem with respect to terminals (a - b) and answer:

- 4) The Norton current is:
a. 2A b. 6A c. 3A d. 4A
- 5) The Norton resistance is:
a. 0.75Ω b. 3Ω c. 1.5Ω d. 6Ω
- 6) The current I is:
a. 1.5A b. 4A c. 3A d. 6A

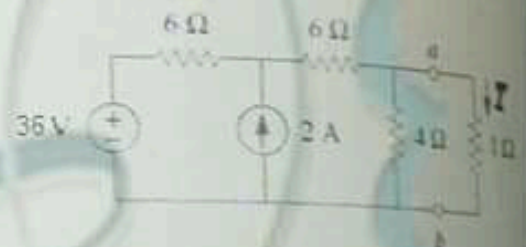


Fig 4

* For the questions 7,8 consider the current in a 5H inductor is $i(t) = 10(1 - 3t)$ A. Assume $v(0) = 0V$. Answer:

- 7) The voltage at the terminals of the inductor at $t = 3s$ is:
a) 150V b) -150V
c) 350V d) -350V
- 8) The Energy stored in the inductor at $t = 1s$ is:
a) -320J b) 250J
c) 1000J d) 500J

* The switch in Figure 5 has been closed for a long time. It opens at $t = 0$

- 9) The steady state response $i(t)$ is
- 10) The transient response $i(t)$ is
- 11) The complete response is
- 12) The inductor current at $t = 200ms$ is

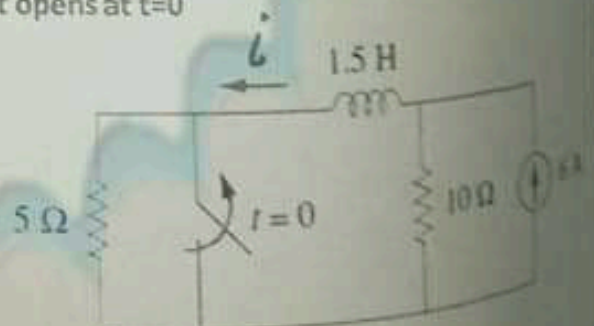


Fig 5

Q1 →

for super mesh →

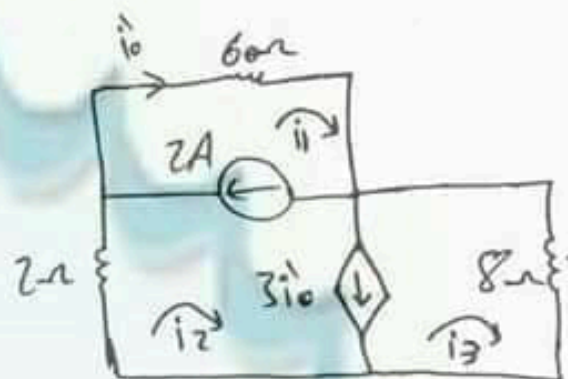
$$6i_1 + 8i_3 + 2i_2 = 0 \quad (1)$$

$$i_1 - i_2 = 2 \quad (2)$$

$$i_2 - i_3 = 3i_0 \quad (3)$$

$$i_0 = i_1 \quad (4)$$

$$\therefore i_0 = -2.5 \text{ A}$$



$$\frac{V_1}{8} + \frac{V_1 - 24}{8} + 3i_0'' = 0 \quad (1)$$

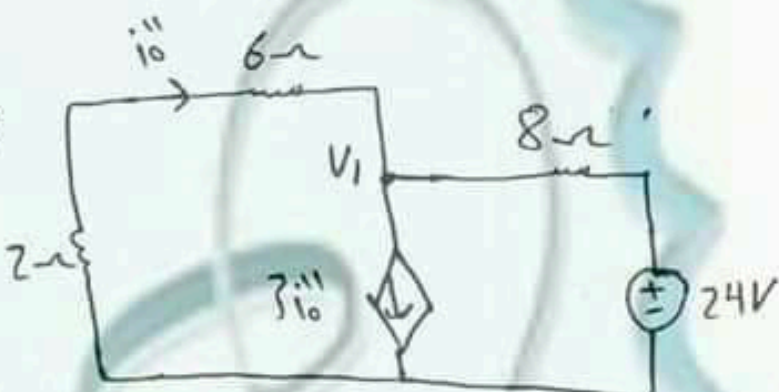
$$i_0'' = \frac{-V_1}{8} \quad (2)$$

$$\therefore V_1 = -24 \text{ V}$$

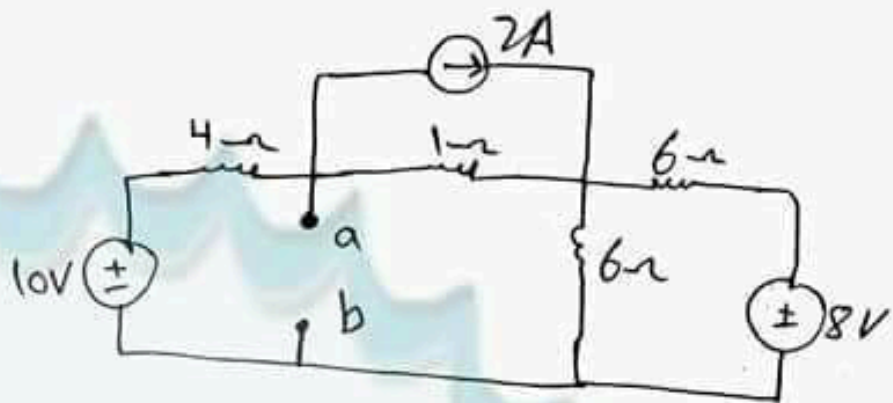
$$i_0'' = 3 \text{ A}$$

$$i_0 = i_0' + i_0'' = -2.5 + 3 = 0.5 \text{ A}$$

$$P_{6\Omega} = i^2 R = 0.5^2 \times 6 = \frac{6}{4} = 1.5 \text{ W}$$



Q2 →

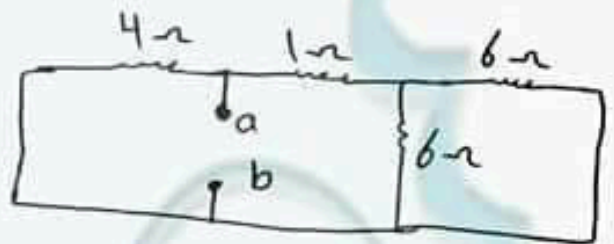


a) $R = R_{th} \rightarrow$

$$6 // 6 = 3 \Omega$$

3 series with 1 = 4Ω

$$4 // 4 = \boxed{2 \Omega}$$

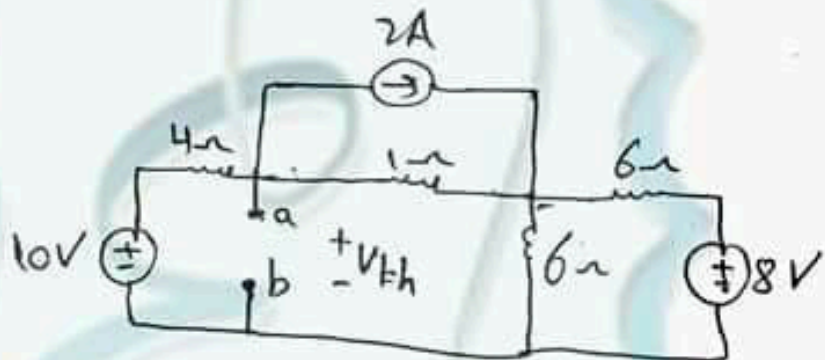


b) $V_{th} \rightarrow$

using ~~super~~ source transform

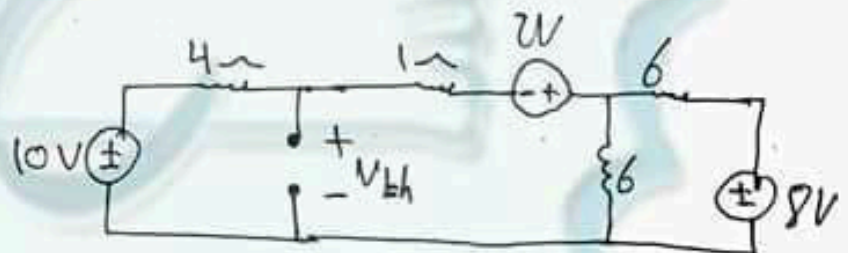
(2A with 1Ω)

(8V with 6Ω)



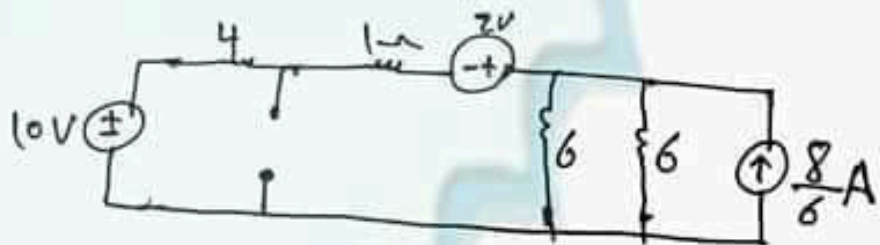
$$6 // 6 = 3$$

($\frac{8}{6}$ A) with 3Ω



$$-10 - 2 + 4 + 8i = 0$$

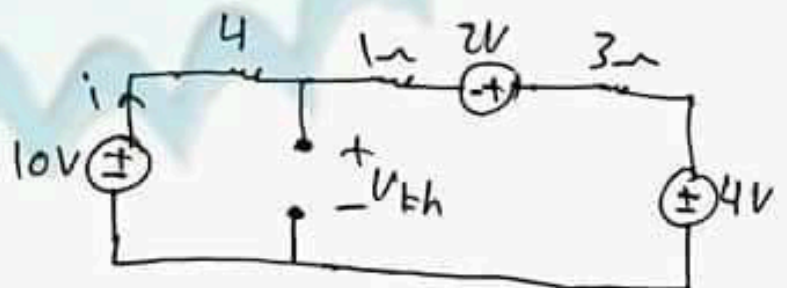
$$\therefore i = 1A$$



$$-V_{th} - 4i + 10 = 0$$

$$\therefore \boxed{V_{th} = 6V}$$

$$P_{max} = \frac{V_{th}^2}{4R_{th}} = \boxed{4.5W}$$



Q3 →

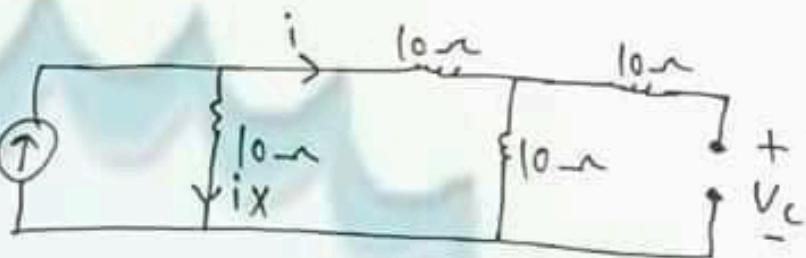
using current divider →

$$i_x = 18 \times \frac{20}{10+20} = \boxed{12A}$$

$$i = 18 - 12 = 6A$$

$$V_c = 10i = \boxed{60V}$$

$$W_c = \frac{1}{2} C V_c^2 = \frac{1}{2} \times \frac{5}{100} \times 3600 = \boxed{90J}$$



Q4 →

$I_N \rightarrow$

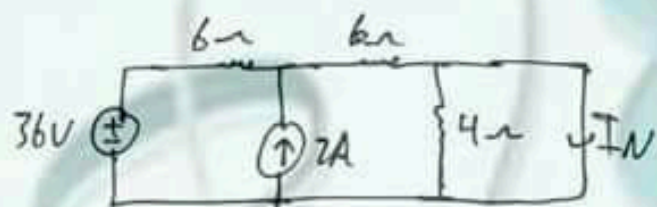
for super mesh →

$$6i_1 + 6i_2 = 36 \rightarrow \boxed{i_1 + i_2 = 6} \quad (1)$$

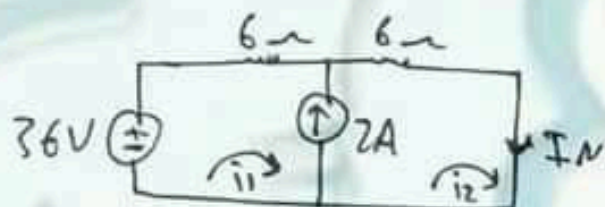
$$\boxed{i_2 - i_1 = 2A} \quad (2)$$

$$i_1 = 2A \quad i_2 = 4A$$

$$\boxed{I_N = i_2 = 4A}$$



4Ω → تحذف



$R_N \rightarrow$

$$6 \text{ series } 6 = 12\Omega$$

$$12 // 4 = \boxed{3\Omega} = R_N$$



$$I = \frac{4 \times 3}{3+1} = \boxed{3A}$$



Q5 $\rightarrow L = 5H \quad i(t) = 10 - 30t \text{ A} \quad V(0) = 0V$

$$V_L = L \frac{di}{dt} = 5 \times \frac{d(10 - 30t)}{dt} = 5 \times -30 = \boxed{-150V}$$

W_L at $t=1 \rightarrow$

$$W_L = \frac{1}{2} Li^2 = \frac{1}{2} \times 5 \times (10 - 30(1))^2 = \frac{1}{2} \times 5 \times 400 = \boxed{1000J}$$

Q6 $\rightarrow$

$$i(0) = \boxed{6A}$$

$$i(\infty) = \frac{6 \times 10}{10 + 5} = \boxed{4A}$$

$$i = i(\infty) + [i(0) - i(\infty)] e^{-\frac{t}{T}}$$

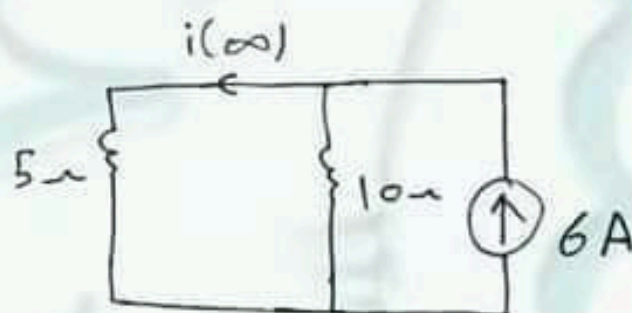
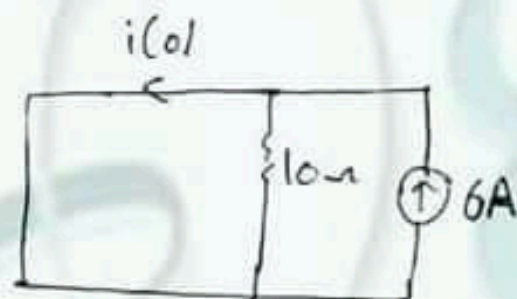
$$T = \frac{L}{R_{eq}}$$

$$R_{eq} = R_{th} = 15\Omega$$

$$\therefore T = \frac{1.5}{15} = 0.1$$

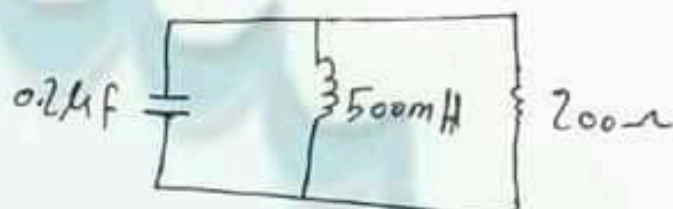
$$\therefore \boxed{i = 4 + 2e^{-10t}} \text{ A}$$

$$i(200ms) = 4 + 2e^{-10(200m)} = 4 + 2e^{-2} = \boxed{4.27A}$$



Q7 $\rightarrow$ find $\alpha, \omega, v(t)$

$$V(0) = 0 \quad I(0) = 2$$



Soln $\alpha = \frac{1}{2RC} = 12500$

$$\omega_0 = \frac{1}{\sqrt{LC}} = 3162.28$$

$\alpha > \omega_0$ (overdamped)

$$V(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

$$s_1 = -\alpha - \sqrt{\alpha^2 - \omega_0^2} = -24593.4$$

$$s_2 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} = -406.6$$

$$V(0) = 0 = A_1 e^0 + A_2 e^0 \Rightarrow A_1 = -A_2 \quad (1)$$

~~$\frac{V}{200} + i + \frac{C \partial V}{\partial t} = 0$~~

$$\frac{V}{200} + i + \frac{C \partial V}{\partial t} = 0$$

$$\frac{V(0)}{200} + i(0) + C \frac{\partial V(0)}{\partial t} = 0 \rightarrow \frac{\partial V(0)}{\partial t} = \frac{-2}{0.2 \mu} = -10 \text{ M}$$

$$\frac{\partial V(t)}{\partial t} = A_1 s_1 e^{s_1 t} + A_2 s_2 e^{s_2 t}$$

$$\frac{\partial V(0)}{\partial t} = A_1 s_1 e^0 + A_2 s_2 e^0 = -10 \text{ M} \quad (2)$$

$$\therefore A_1 = 413.45 \quad A_2 = -413.45$$

$$\therefore V(t) = 413.45 e^{-24593.4 t} - 413.45 e^{-406.6 t} \quad \text{V}$$



15/5/2018

أسئلة سنوات سابقة

المادة: سيركت 1

Choose the correct answer.

* For the question 1,2,3,4 Consider the circuit shown in Figure 1 and answer:

1) The value of (V_1) is:

- a) 11.6 v b) 22 v
c) -22 v d) 7.4 v

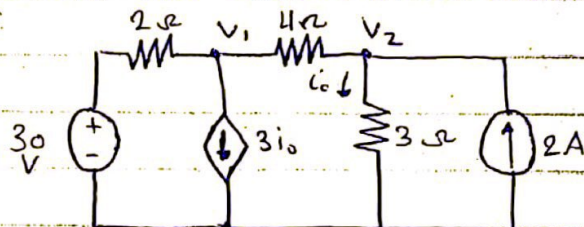


Fig. 1

2) The value of the current (i_o) is:

- a) 2 A b) 2.8 A
c) 1.4 A d) -1.4 A

3) The value of the voltage (V_2) is:

- a) -4.8 V b) 4 V
c) -2.4 v d) 8.4 v

4) The power absorbed by 4Ω resistor is:

- a) 6.4 w b) 3.64 w
c) 2.56 w d) 10 w

* For the questions 5,6,7 consider the circuit shown in figure 2:

5) The value of R_{th} with respect to terminals ab is:

- a) 3Ω b) 6Ω
c) 2Ω d) 17Ω

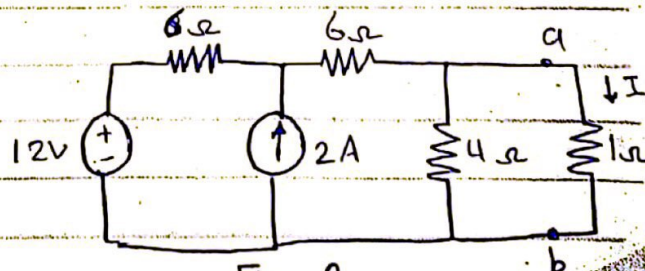


Fig. 2

Don't miss the change ... Be part of it

1



أسئلة سنوات سابقة

المادة:

6) The value of V_{TH} from the terminals (a,b) is:

- a) 3 V b) 6 V
c) 9 V d) 12 V

7) The power delivered to 1Ω resistor is:

- a) 2.25 W b) 1.5 W
c) 4 W d) 4.45 W

* For the questions 8, 9, 10, 11 consider the circuit shown in Figure 3 and answer:

8) The value of $v(t=\infty)$ is:

- a) 15 V b) -15 V
c) 20 V d) -20 V

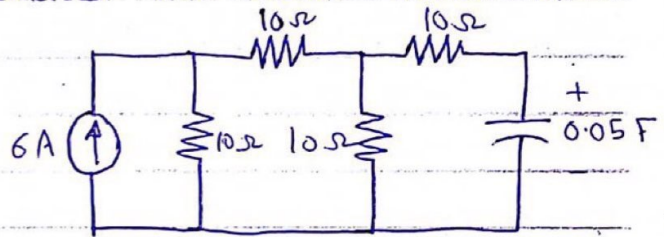


Fig. 3

9) The value of $i(t=\infty)$ is:

- a) 2 A b) 10 A
c) 4.68 A d) 0 A

b) The value of τ (time constant) is:

- a) 1.2 s b) 833 ms
c) 1.6 s d) 8.6 s

11) If the initial value of the capacitor voltage is $v(0) = 10V$. The initial value of the capacitor current $i(0)$ is:

- a) 0.6 A b) 1.2 A
c) 10.63 A d) 5.6 A



أسئلة سنوات سابقة

المادة:

* If the voltage across a 1-mF capacitor is $v(t) = 60 \cos 100t$ mV
answer 12, 13 :

12) The current through the capacitor is:

- a) $-6 \sin(100t)$ mA b) $6 \cos(100t)$ mA
c) $-12 \sin(100t)$ mA d) $-6 \cos(100t)$ mA

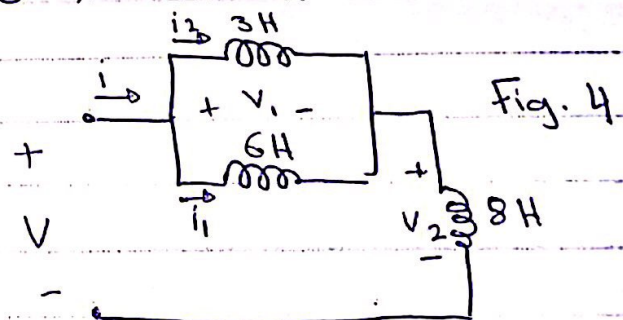
13) The energy stored is:

- a) $1.8 \sin^2(100t)$ μ J b) $1.8 \cos^2(100t)$ μ J
c) $1800 \sin^2(100t)$ mJ d) $1800 \cos^2(100t)$ mJ

* For the questions 14, 15, 16, 17 consider the circuit shown in figure 4 ($i_1(t) = 0.6 e^{-2t}$ A . If $i_1(0) = 1.4$ A) :

14) The current $i_2(0)$ is:

- a) 0.8 A b) 1.6 A
c) 6 A d) 3.2 A



15) The total current $i_2(t)$ is:

- a) $(-0.8 + 1.4 e^{-2t})$ A b) $(-0.4 + 1.4 e^{-2t})$ A
c) $(-0.4 + 1.2 e^{-2t})$ A d) $(-0.8 + 1.2 e^{-2t})$ A

16) The voltage $v(t)$ is:

- a) $(6 + 1.6 e^{-2t})$ V b) $(1.8 e^{-2t})$ V
c) $(6 + 1.6 e^{+2t})$ V d) $-36 e^{-2t}$ V

17) The voltage $V_1(t)$ is:

- a) $(6 - 1.6 e^{-2t})$ V b) $(6 + 1.6 e^{+2t})$ V
c) $-7.2 e^{-2t}$ V d) $(8 + 1.6 e^{-2t})$ V



أسئلة سنوات سابقة

المادة:

* For the questions 18,19 Refer to Figure 5:

18) the value of the equivalent resistance is:

- a) 9Ω b) 4.5Ω
c) 6Ω d) 4Ω

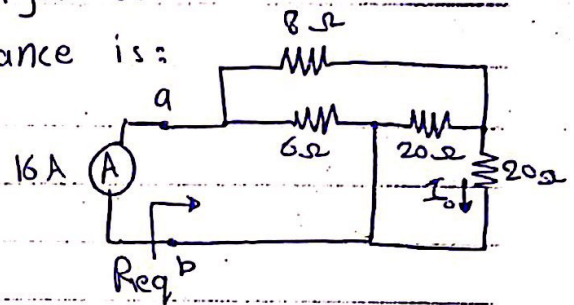


Fig. 5

19) The value of the current i_o is:

- a) 2A b) 4A
c) 4.5A d) 8A

* For the questions 20,21 consider the circuit shown in Figure 6:

20) The voltage across 10Ω resistor is:

- a) 4V b) 50V
c) 40V d) 45V

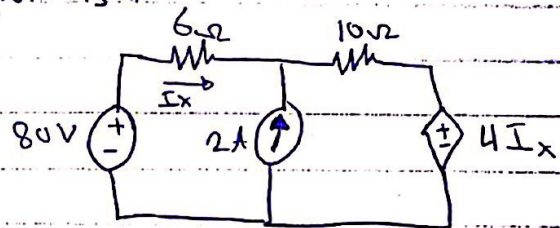


Fig. 6

21) The current I_x is:

- a) 5A b) 2.4A c) -2.4A d) 3A

* For the questions 22,23,24 (consider the circuit shown in Figure 7. Assuming $v(0) = 5V$, $i(0) = 0$, $L = 1H$, $C = 10mF$. And $R = 1.923\Omega$)

22) The value of α (damping factor) is:

- a) $13s^{-1}$ b) $52s^{-1}$ c) $26s^{-1}$ d) $54s^{-1}$

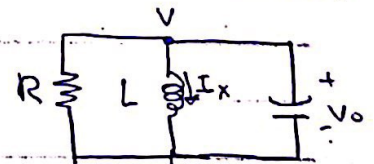


Fig. 7

23) The value of ω_0 (resonant frequency) is:

- a) $52s^{-1}$ b) $10s^{-1}$ c) $20s^{-1}$ d) $26s^{-1}$

24) The general natural response $v(t)$ for $t > 0$ is:

25) A heater draws 10A when connected to 240V supply. How long does it take to consume 480 kJ.

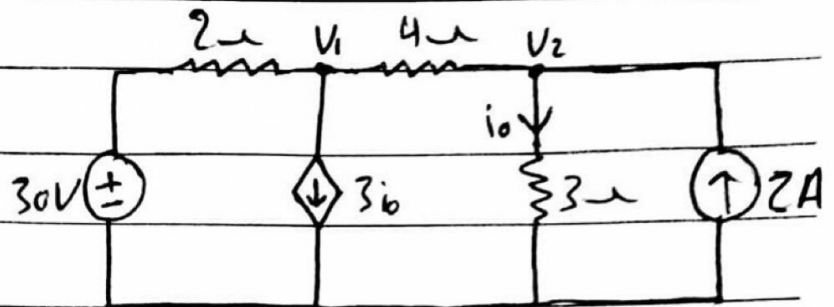
- a) 260ms b) 5ms
c) 20s d) 200s

Don't miss the change ... Be part of it

1, 2, 3, 4 →

soln

at node 1 →



$$\frac{V_1 - 30}{2} + \frac{V_1 - V_2}{4} + 3i_b = 0$$

$$0.75 V_1 - 0.25 V_2 + 3i_b = 15 \quad \text{--- (1)}$$

at node 2 →

$$\frac{V_2 - V_1}{4} + \frac{V_2}{3} = 2 \rightarrow -0.25 V_1 + 0.58 V_2 = 2 \quad \text{--- (2)}$$

$$\text{from the circuit} \rightarrow i_o = \frac{V_2}{3} \quad \text{--- (3)}$$

$$\therefore V_1 = 11.566 \text{ V} \quad V_2 = 8.43 \quad i_o = 2.81 \text{ A}$$

$$\text{the power in } (4\Omega) = i^2 R = \left(\frac{V_1 - V_2}{4} \right)^2 \times R$$

$$\therefore P = \left(\frac{11.566 - 8.43}{4} \right)^2 \times 4 = 2.46 \text{ W}$$

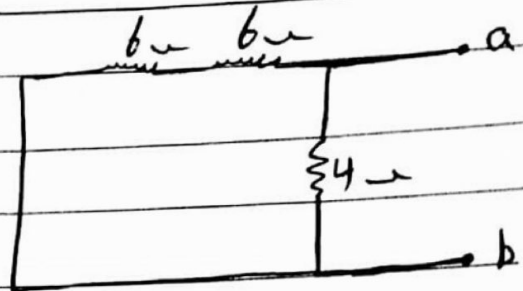
∴ 1) a 2) b 3) d 4) c

5, 6, 7 →

* to find R_{th} →

Voltage source → open

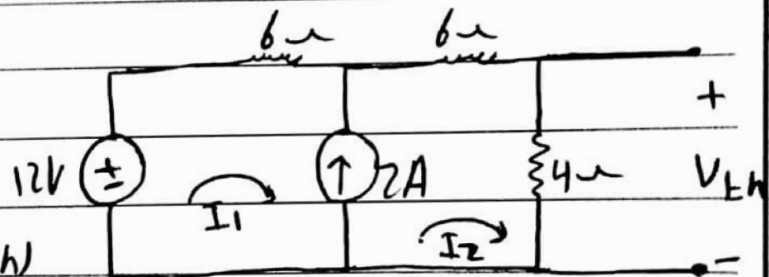
current source → short

 6 series with $6 = 12 \Omega$

$$12 // 4 = 3 \Omega = R_{th}$$

* to find V_{th} →

$$V_{th} = V_{4\Omega}$$



at loop 22 → (super mesh)

$$-12 + 6I_1 + 6I_2 + 4I_2 = 0$$

$$\therefore 6I_1 + 10I_2 = 12 \quad \text{--- (1)}$$

$$I_2 - I_1 = 2 \quad \text{--- (2)}$$

$$\therefore I_1 = -0.5A \quad I_2 = 1.5A$$

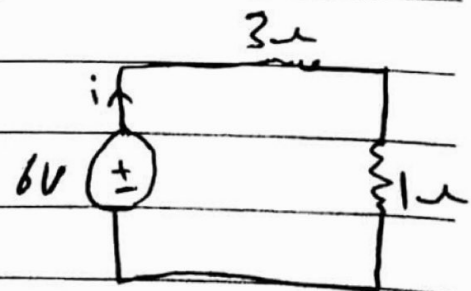
~~at loop 11~~

$$* V_{th} = V_{4\Omega} = 4I_2 = 4 \times 1.5 = 6V$$

the power in (1Ω) →

$$-6 + 4i = 0 \rightarrow i = 1.5A$$

$$P = i^2 R = (1.5)^2 \times 1 = 2.25W$$



∴ 5) a 6) b 7) a

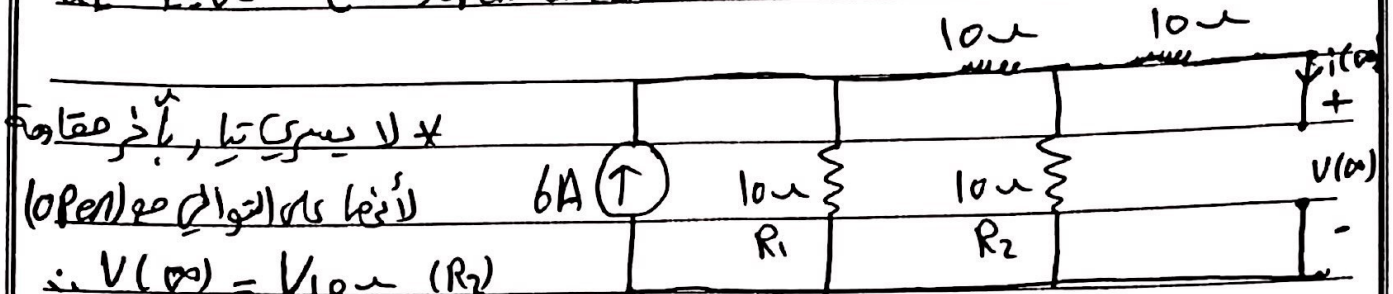
Don't miss the change..

Be part of it !!

2



8, 9, 10 →

at $t = \infty$ $C \rightarrow$ open circuit

$$\therefore V(\infty) = V_{10\Omega} (R_2)$$

$$10 \text{ series with } 10 = 20\Omega \quad 20 // 10 = \frac{20}{3}\Omega$$

$$V_{\frac{20}{3}} = iR = 6 \times \frac{20}{3} = 40V = V_{R_1} = V_{20\Omega}$$

$$i_{20\Omega} = \frac{V}{R} = \frac{40}{20} = 2A \quad \therefore V_{R_2} = V(\infty) = iR = 10 \times 2 = 20V$$

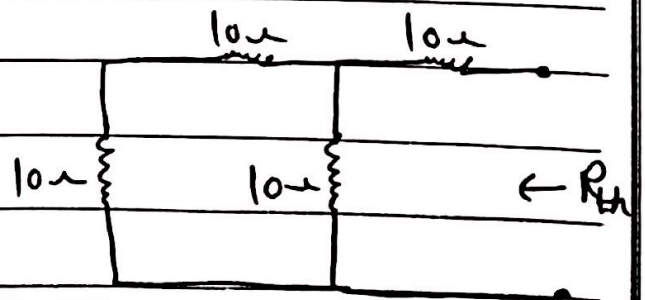
$$i(\infty) = 0A \quad (\text{open})$$

$$T = R_{eq} \times C, \quad R_{eq} = R_{th} \quad (\text{مقاومة طرفي المقاومة})$$

$$10 \text{ series with } 10 = 20\Omega$$

$$20 // 10 = \frac{20}{3}\Omega$$

$$R_{th} = \frac{20}{3} + 10 = \frac{50}{3}\Omega$$



$$T = R_{eq} \times C = \frac{50}{3} \times 0.05 = 833ms$$

8) c

9) d

10) b

Don't miss the change.
Be part of it !!

3

Red

11, 12, 13 →

11) $V(0) = 10V$ $i(0) = ??$

(step response) →

بسیب وجود صفر غیر صفری

$$V(t) = V(\infty) + [V(0) - V(\infty)] e^{-\frac{t}{\tau}}$$

$$V(t) = 20 + [10 - 20] e^{-1.2t} = 20 - 10e^{-1.2t}$$

$$i = C \frac{\partial V}{\partial t} = 0.05 \times \frac{\partial (20 - 10e^{-1.2t})}{\partial t}$$

$$\therefore i = 0.05 \times 1.2 \times 10 e^{-1.2t} \quad \therefore i(t) = 0.6e^{-1.2t} A$$

$$i(0) = 0.6A \quad \therefore \textcircled{a}$$

$$12) i = C \frac{\partial V}{\partial t} = 1 \times 10^{-3} \times \frac{\partial (60 \cos(100t) m)}{\partial t}$$

$$\therefore i = 1 \times 10^{-3} \times 60 \times 100 \times -\sin(100t) m = -6 \sin(100t) mA \quad \therefore \textcircled{a}$$

$$13) W = \frac{1}{2} C V^2 = \frac{1}{2} \times 1 \times 10^{-3} \times (60 \cos(100t) m)^2$$

$$\therefore W = \frac{1}{2} \times 10^{-3} \times 3600 \times \cos^2(100t) J = 1.8 \cos^2(100t) J$$

 $\therefore \textcircled{b}$

Don't miss the change...

Be part of it !!

4

V.P.S.

14, 15, 16, 17 →

$$i_1(t) = 0.6 e^{-2t} \quad \therefore i_1(0) = 0.6 \text{ A} \quad i(0) = 1.4 \text{ A}$$

$$i(0) = i_1(0) + i_2(0) \quad \therefore \boxed{i_2(0) = 0.8 \text{ A}}$$

$$V_1(t) = V_{6H} = V_{3H} = L \frac{di_1(t)}{dt} = 6 \times \frac{d(0.6 e^{-2t})}{dt}$$

$$\therefore V_1(t) = 6 \times 0.6 \times -2 e^{-2t} = \boxed{-7.2 e^{-2t} \text{ V}}$$

$$i_2(t) = \frac{1}{L} \int_{t_0}^t V_1 dt + i_2(0) \quad \therefore i_2(t) = \frac{1}{3} \int_0^t -7.2 e^{-2t} dt + 0.8$$

$$\therefore i_2(t) = \frac{1}{3} \left[\frac{-7.2}{-2} e^{-2t} \right]_0^t + 0.8 = 1.2 [e^{-2t} - 1] + 0.8$$

$$\therefore \boxed{i_2(t) = -0.4 + 1.2 e^{-2t} \text{ A}} \quad \boxed{i = i_1 + i_2 = -0.4 + 1.8 e^{-2t} \text{ A}}$$

the equivalent inductance is → $(3/6) + 8 = 10 \text{ H}$

the total current is → $i = -0.4 + 1.8 e^{-2t} \text{ A}$

$$\therefore V(t) = L \frac{di}{dt} = 10 \frac{d(-0.4 + 1.8 e^{-2t})}{dt} = 10 \times 1.8 \times -2 \times e^{-2t}$$

$$\therefore \boxed{V(t) = -36 e^{-2t} \text{ V}}$$

$\therefore$ 14) a 15) c 16) d 17) c

Don't miss the change:
Be part of it !!

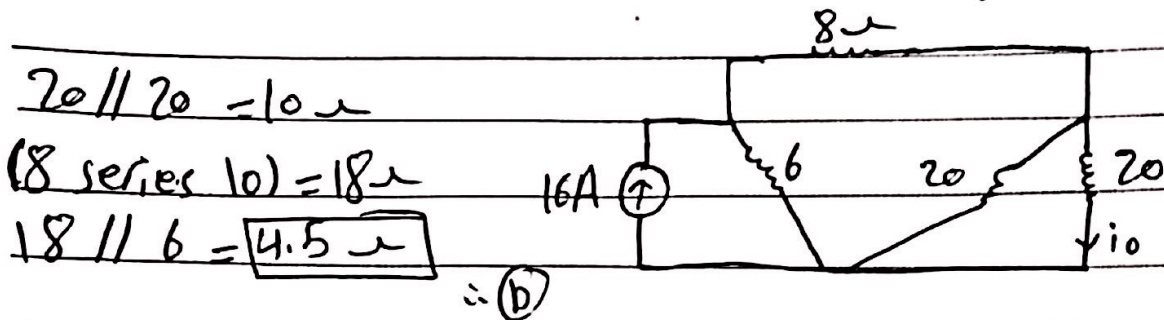
5

YES

18, 19 →

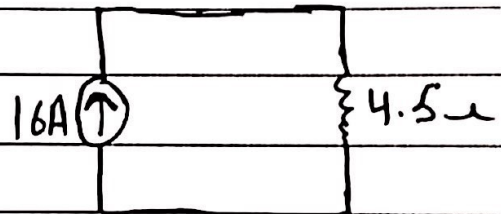
18) Req ??

نقوم بتفسير شكل الدارة للتبسيط ←



$$19) V_{4.5} = iR = 16 \times 4.5 = 72V$$

$$V_6 = V_{18} = 72V \text{ (parallel)}$$



$$i_{18} = \frac{V}{R} = \frac{72}{18} = 4A = i_{10} = i_8 \text{ (series)}$$

$$V_{10} = iR = 4 \times 10 = 40V = V_{20} = V_{20} \text{ (parallel)}$$

$$i_o = \frac{V_{20}}{R} = \frac{40}{20} = 2A \quad \therefore (a)$$

SOL: EXAM

Circuit 1 "find"

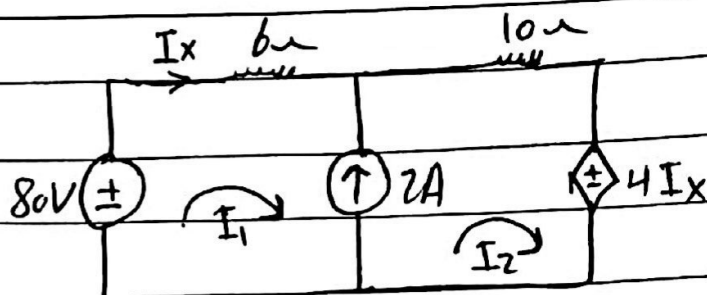
PERIOD

15/5/2018

NOTE

$20, 21 \rightarrow$

(Super mesh) $\rightarrow$



$$-80 + 6I_1 + 10I_2 + 4I_x = 0 \quad (1)$$

$$I_2 - I_1 = 2 \quad (2)$$

$$I_x = I_1 \quad (3)$$

$$\therefore I_1 = 3A$$

$$I_2 = 5A$$

$$I_x = 3A$$

$$V_{10\Omega} = I_2 R = 5 \times 10 = 50V$$

$\therefore$ ~~20) b~~ 21) d

Don't miss the change..
Be part of it !!



SOL: EXAM circuit 1 "final"

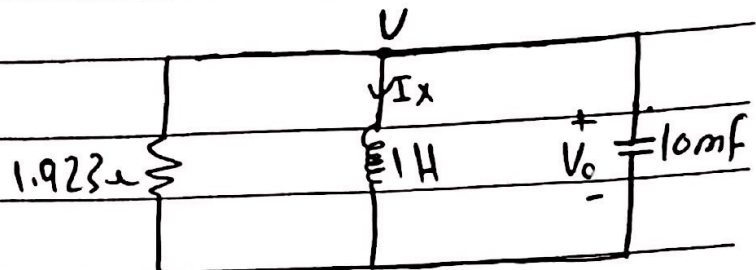
PERIOD 15/5/2018

NOTE

22, 23, 24, 25 →

$V(0) = 5V$ $i(0) = 0A$

soln



22) $\alpha = \frac{1}{2RC} = 26$ (c)

23) $\omega_0 = \frac{1}{\sqrt{LC}} = 10$ (b)

24) $\alpha > \omega_0$ (over damped)

$\therefore V(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$ $\therefore$ (a)

~~5.4.2018~~

25) $I = 10A$ $V = 240V$ $W = 480kJ$ $t = ??$

$P = IV = 10 \times 240 = 2400W$

$P = \frac{\partial W}{\partial t} \rightarrow 2400 = \frac{480kJ}{t} \rightarrow \boxed{t = 200s}$

Don't miss the change.
Be part of it !!

8

100%